

ACADEMIC REGULATIONS
COURSE STRUCTURE & DETAILED SYLLABUS
CHOICE BASED CREDIT SYSTEM
R22

Bachelor of Technology (B.Tech)

Aeronautical Engineering

B. Tech. - Regular Four Year Degree Programme
(For batches admitted from the academic year 2022-2023)
&
(For batches admitted Lateral Entry Scheme from the
academic year 2023-2024)



MLR Institute of Technology

(Autonomous)

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FOREWORD

The autonomy is conferred on MLR Institute of Technology by UGC, based on its performance as well as future commitment and competency to impart quality education. It is a mark of its ability to function independently in accordance with the set norms of the monitoring bodies like UGC and AICTE. It reflects the confidence of the UGC in the autonomous institution to uphold and maintain standards it expects to deliver on its own behalf and thus awards degrees on behalf of the college. Thus, an autonomous institution is given the freedom to have its own **curriculum, examination system and monitoring mechanism**, independent of the affiliating University but under its observance.

MLR Institute of Technology is proud to win the credence of all the above bodies monitoring the quality in education and has gladly accepted the responsibility of sustaining, if not improving upon the standards and ethics for which it has been striving for more than a decade in reaching its present standing in the arena of contemporary technical education. As a follow up, statutory bodies like Academic Council and Boards of Studies are constituted with the guidance of the Governing Body of the College and recommendations of the JNTU Hyderabad to frame the regulations, course structure and syllabi under autonomous status.

The autonomous regulations, course structure and syllabi have been prepared after prolonged and detailed interaction with several expertise solicited from academics, industry and research, in accordance with the vision and mission of the college in order to produce quality engineering graduates to the society.

All the faculty, parents and students are requested to go through all the rules and regulations carefully. Any clarifications, if needed, are to be sought, at appropriate time with principal of the college, without presumptions, to avoid unwanted subsequent inconveniences and embarrassments. The Cooperation of all the stake holders is sought for the successful implementation of the autonomous system in the larger interests of the college and brighter prospects of engineering graduates.

PRINCIPAL

ACADEMIC REGULATIONS (R22)

B. Tech. - Regular Four Year Degree Programme (For batches admitted from the academic year 2022-23)

1.0 For pursuing four year Under Graduate Degree Programme of study in Engineering & Technology (UGP in E&T) offered by MLR Institute of Technology under Autonomous status is herein referred to as MLRIT (Autonomous):

All the rules specified herein approved by the Academic Council will be in force and applicable to students admitted from the Academic Year 2022-23 onwards. Any reference to “Institute” or “College” in these rules and regulations shall stand for M L R Institute of Technology (Autonomous).

All the rules and regulations, specified hereafter shall be read as a whole for the purpose of interpretation as and when a doubt arises, the interpretation of the Chairman, Academic Council is final. As per the requirements of statutory bodies, the Principal, M L R Institute of Technology shall be the chairman Academic Council.

2.0 Eligibility for Admission

2.1 Admission to the undergraduate(UG) programme shall be made either on the basis of the merit rank obtained by the qualified student in entrance test conducted by the Telangana State Government (EAMCET) or the College or on the basis of any other order of merit approved by the University/TSHE, subject to reservations as prescribed by the government from time to time.

2.2 The medium of instructions for the entire undergraduate programme in Engineering & Technology will be **English** only.

3.0 B.Tech. Programme Structure

3.1 A student after securing admission shall complete the B.Tech. programme in a minimum period of **four** academic years (8 semesters), and a maximum period of **eight** academic years (16 semesters) starting from the date of commencement of first year first semester, failing which student shall forfeit seat in B.Tech course. Each student shall secure 160 credits (with CGPA ≥ 5) required for the completion of the undergraduate programme and award of the B.Tech. degree.

3.2 UGC/ AICTE specified definitions/ descriptions are adopted appropriately for various terms and abbreviations used in these academic regulations/ norms, which are listed below.

3.2.1 Semester Scheme

Each undergraduate programme is of 4 academic years (8 semesters) with the academic year divided into two semesters of 22 weeks (≥ 90 instructional days) each and in each semester - ‘Continuous Internal Evaluation (CIE)’ and ‘Semester End Examination (SEE)’ under Choice Based Credit System (CBCS) and Credit Based Semester System (CBSS) indicated by UGC, and curriculum/course structure suggested by AICTE are followed.

3.2.2 Credit Courses

All subjects/ courses are to be registered by the student in a semester to earn credits which shall be assigned to each subject/ course in an L: T: P: C (lecture periods: tutorial periods: practical periods: credits) structure based on the following general pattern.

- One credit for one hour/ week/ semester for Theory/ Lecture (L) courses or Tutorials.
- One credit for two hours/ week/ semester for Laboratory/ Practical (P) courses.

Courses like Environmental Science, Constitution of India, Intellectual Property Rights, and Gender Sensitization Lab are mandatory courses. These courses will not carry any credits.

3.2.3 Subject Course Classification

All subjects/ courses offered for the undergraduate programme in E&T (B.Tech. degree programmes) are broadly classified as follows. The College has followed almost all the guidelines issued by AICTE/UGC.

S. No.	Broad Course Classification	Course Group/ Category	Course Description
1	Foundation Courses (FnC)	BS–Basic Sciences	Includes Mathematics, Physics and Chemistry Subjects.
2		ES–Engineering Sciences	Includes Fundamental Engineering Subjects.
3		HS–Humanities and Social Sciences	Includes subjects related to Humanities, Social Sciences and Management.
4	Core Courses (CoC)	PC–Professional Core.	Includes core subjects related to the parent discipline/ department/ branch of Engineering.
5	Elective Courses (ElC)	PE–Professional Electives	Includes elective subjects related to the parent discipline/ department/ branch of Engineering.
6		OE–Open Electives	Elective subjects which include interdisciplinary subjects or subjects in an area outside the parent discipline/ department/ branch of Engineering.
7	Core Courses	Project Work	B.Tech. Project or UG Project or UG Major Project or Project Stage I & II.
8		Industry Training/ Internship / Industry Oriented Mini- project/ Mini- Project/ Skill Development Courses.	Industry Training/ Internship/ Industry Oriented Mini-Project/ Mini-Project/ Skill Development Courses.
9		Seminar	Seminar/ Colloquium based on core contents related to parent discipline/ department/ branch of Engineering.
10	Minor Courses	-	1 or 2 Credit Courses (subset of HS)
11	Mandatory Courses (MC)	-	Mandatory Courses (non-credit)

4.0 Course Registration

- 4.1 A 'faculty advisor or counselor' shall be assigned to a group of 20 students, who will advise the students about the undergraduate programme, its course structure and curriculum, choice/option for subjects/ courses, based on their competence, progress, pre-requisites and interest.
- 4.2 The academic section of the college invites 'registration forms' from students before the beginning of the semester through 'on-line registration', ensuring 'date and time stamping'. The on-line registration requests for any 'current semester' shall be **completed before the commencement of SEEs (Semester End Examinations) of the 'preceding semester'**.
- 4.3 A student can apply for **on-line** registration, **only after** obtaining the '**written approval**' from faculty advisor/counselor, which should be submitted to the college academic section through the Head of the Department. A copy of it shall be retained with the Head of the Department, Faculty Advisor/ Counselor and the student.
- 4.4 A student may be permitted to register for all the subjects/ courses in a semester as specified in the course structure with maximum additional subject(s)/course(s) limited to 6 Credits (any 2 elective subjects), based on **progress** and SGPA/ CGPA, and completion of the '**pre-requisites**' as indicated for various subjects/ courses, in the department course structure and syllabus contents.
- 4.5 Choice for '**additional subjects/ courses**', not more than any 2 elective subjects in any Semester, must be clearly indicated, which needs the specific approval and signature of the Faculty Advisor/Mentor/HOD.
- 4.6 If the student submits ambiguous choices or multiple options or erroneous entries during **on-line** registration for the subject(s) / course(s) under a given/ specified course group/ category as listed in the course structure, only the first mentioned subject/ course in that category will be taken into consideration.
- 4.7 Subject/ course options exercised through **on-line** registration are final and **cannot** be changed or inter-changed; further, alternate choices also will not be considered. However, if the subject/ course that has already been listed for registration by the Head of the Department in a semester could not be offered due to any inevitable or unexpected reasons, then the student shall be allowed to have alternate choice either for a new subject (subject to offering of such a subject), or for another existing subject (subject to availability of seats). Such alternate arrangements will be made by the Head of the Department, with due notification and time-framed schedule, within **a week** after the commencement of class-work for that semester.
- 4.8 Dropping of subjects/ courses may be permitted, only after obtaining prior approval from the faculty advisor/ counselor 'within a period of 15 days' from the beginning of the current semester.
- 4.9 **Open Electives:** The students have to choose three Open Electives (OE-I, II & III) from the list of Open Electives given by other departments. However, the student can opt for an Open Elective subject offered by his own (parent) department, if the student has not registered and not studied that subject under any category (Professional Core, Professional Electives, Mandatory Courses etc.) offered by parent department in any semester. Open Elective subjects already studied should not repeat/should not match with any category (Professional Core, Professional Electives, Mandatory Courses etc.) of subjects even in the forthcoming semesters.

4.10 Professional Electives: The students have to choose six Professional Electives (PE-I to VI) from the list of professional electives given.

5.0 Subjects/ courses to be offered

5.1 A subject/ course may be offered to the students, **only if** a minimum of 15 students opt for it.

5.2 More than **one faculty member** may offer the **same subject** (lab/ practical may be included with the corresponding theory subject in the same semester) in any semester. However, selection of choice for students will be based on - '**first come first serve** basis and CGPA criterion' (i.e. the first focus shall be on early **on-line entry** from the student for registration in that semester, and the second focus, if needed, will be on CGPA of the student).

5.3 If more entries for registration of a subject come into picture, then the Head of the Department concerned shall decide, whether or not to offer such a subject/ course for **two (or multiple) sections**.

5.4 In case of options coming from students of other departments/ branches/ disciplines (not considering **open electives**), first **priority** shall be given to the student of the '**parent department**'.

6.0 Attendance requirements:

6.1 A student shall be eligible to appear for the semester end examinations, if the student acquires a minimum of 75% of attendance in aggregate of all the subjects/ courses (including attendance in mandatory courses like Environmental Science, Constitution of India, Intellectual Property Rights, and Gender Sensitization Lab) for that semester. **Two periods** of attendance for each theory subject shall be considered, if the student appears for the mid-term examination of that subject.

6.2 Shortage of attendance in aggregate upto 10% (65% and above, and below 75%) in each semester may be condoned by the college academic committee on genuine and valid grounds, based on the student's representation with supporting evidence.

6.3 A stipulated fee shall be payable for condoning of shortage of attendance.

6.4 Shortage of attendance below 65% in aggregate shall in **NO** case be condoned.

6.5 **Students whose shortage of attendance is not condoned in any semester are not eligible to take their end examinations of that semester. They get detained and their registration for that semester shall stand cancelled**, including all academic credentials (internal marks etc.) of that semester. **They will not be promoted to the next semester.** They may seek re-registration for all those subjects registered in that semester in which the student is detained, by seeking re-admission into that semester as and when offered; if there are any professional electives and/ or open electives, the same may also be re-registered if offered. However, if those electives are not offered in later semesters, then alternate electives may be chosen from the **same** set of elective subjects offered under that category.

6.6 A student fulfilling the attendance requirement in the present semester shall not be eligible for readmission into the same class.

7.0 Academic Requirements

The following academic requirements have to be satisfied, in addition to the attendance requirements mentioned in Item No. 6.

7.1 A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/ course, if student secures not less than 35% (21 marks out of 60 marks) in the semester end examination, and a minimum of 40% (40 marks out of 100 marks) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together; in terms of letter grades, this implies securing 'C' grade or above in that subject/ course.

7.2 A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to Real-time Research Project (or) Field Based Research Project (or) Industry Oriented Mini Project (or) Internship (or) Seminar, if the student secures not less than 40% marks (i.e. 40 out of 100 allotted marks) in each of them. The student is deemed to have failed, if he (i) does not submit a report on Industry Oriented Mini Project/Internship, or (ii) not make a presentation of the same before the evaluation committee as per schedule, or (iii) secures less than 40% marks in Real-time Research Project (or) Field Based Research Project (or) Industry Oriented Mini Project (or) Internship evaluations.

A student may reappear once for each of the above evaluations, when they are scheduled again; if the student fails in such 'one reappearance' evaluation also, the student has to reappear for the same in the next subsequent semester, as and when it is scheduled.

7.3 Promotion Rules

S. No.	Promotion	Conditions to be fulfilled
1	First year first semester to first year second semester	Regular course of study of first year first semester.
2	First year second semester to Second year first semester	(i) Regular course of study of first year second semester. (ii) Must have secured at least 20 credits out of 40 credits i.e., 50% credits up to first year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.
3.	Second year first semester to Second year second semester	Regular course of study of second year first semester.
4	Second year second semester to Third year first semester	(i) Regular course of study of second year second semester. (ii) Must have secured at least 48 credits out of 80 credits i.e., 60% credits up to second year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.
5	Third year first semester to Third year second semester	Regular course of study of third year first semester.

6	Third year second semester to Fourth year first semester	(i) Regular course of study of third year second semester. (ii) Must have secured at least 72 credits out of 120 credits i.e., 60% credits up to third year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.
7	Fourth year first semester to Fourth year second semester	Regular course of study of fourth year first semester.

- 7.4 A student (i) shall register for all courses/subjects covering 160 credits as specified and listed in the course structure, (ii) fulfills all the attendance and academic requirements for 160 credits, (iii) earn all 160 credits by securing $SGPA \geq 5.0$ (in each semester), and $CGPA \geq 5$ (at the end of 8 semesters), (iv) **passes all the mandatory courses**, to successfully complete the undergraduate programme. The performance of the student in these 160 credits shall be considered for the calculation of the final CGPA **(at the end of undergraduate programme)**, and shall be indicated in the grade card / marks memo of IV-year II semester.
- 7.5 If a student registers for ‘**extra subjects**’ (in the parent department or other departments/branches of Engg.) other than those listed subjects totaling to 160 credits as specified in the course structure of his department, the performances in those ‘**extra subjects**’ (although evaluated and graded using the same procedure as that of the required 160 credits) will not be considered while calculating the SGPA and CGPA. For such ‘**extra subjects**’ registered, percentage of marks and letter grade alone will be indicated in the grade card / marks memo as a performance measure, subject to completion of the attendance and academic requirements as stated in regulations Items 6 and 7.1 – 7.4 above.
- 7.6 A student eligible to appear in the semester end examination for any subject/ course, but absent from it or failed (thereby failing to secure ‘**C**’ grade or above) may reappear for that subject/ course in the supplementary examination as and when conducted. In such cases, internal marks (CIE) assessed earlier for that subject/ course will be carried over, and added to the marks to be obtained in the SEE supplementary examination for evaluating performance in that subject.
- 7.7 A student **detained in a semester due to shortage of attendance may be re-admitted in the same semester in the next academic year for fulfillment of academic requirements**. The academic regulations under which a student has been re-admitted shall be applicable. Further, no grade allotments or SGPA/ CGPA calculations will be done for the entire semester in which the student has been detained.
- 7.8 A student **detained due to lack of credits, shall be promoted to the next academic year only after acquiring the required number of academic credits**. The academic regulations under which the student has been readmitted shall be applicable to him.

8.0 Evaluation - Distribution and Weightage of Marks

8.1 The performance of a student in every subject/course (including practicals and Project Stage – I & II) will be evaluated for 100 marks each, with 40 marks allotted for CIE (Continuous Internal Evaluation) and 60 marks for SEE (Semester End-Examination).

8.2 In CIE, for theory subjects, during a semester, there shall be two mid-term examinations. Each Mid-Term examination consists of two parts i) **Part – A** for 10 marks, ii) **Part – B** for 20 marks with a total duration of 2 hours as follows:

1. Mid Term Examination for 30 marks:
 - a. Part - A : Objective/quiz paper for 10 marks.
 - b. Part - B : Descriptive paper for 20 marks.

The objective/quiz paper is set with multiple choice, fill-in the blanks and match the following type of questions for a total of 10 marks. The descriptive paper shall contain 6 full questions out of which, the student has to answer 4 questions, each carrying 5 marks. The **average of the two Mid Term Examinations** shall be taken as the final marks for Mid Term Examination (for 30 marks).

The remaining 10 marks of Continuous Internal Evaluation are distributed as:

2. Assignment for 5 marks. (**Average of 2 Assignments** each for 5 marks)
3. Subject Viva-Voce/PPT/Poster Presentation/ Case Study on a topic in the concerned subject for 5 marks.

While the first mid-term examination shall be conducted on 50% of the syllabus, the second mid-term examination shall be conducted on the remaining 50% of the syllabus.

Five (5) marks are allocated for assignments (as specified by the subject teacher concerned). The first assignment should be submitted before the conduct of the first mid-term examination, and the second assignment should be submitted before the conduct of the second mid-term examination. The average of the two assignments shall be taken as the final marks for assignment (for 5 marks).

Subject Viva-Voce/PPT/Poster Presentation/ Case Study on a topic in the subject concerned for 5 marks before II Mid-Term Examination.

- The Student, in each subject, shall have to earn 35% of marks (i.e. 14 marks out of 40 marks) in CIE, 35% of marks (i.e. 21 marks out of 60) in SEE and Over all 40% of marks (i.e. 40 marks out of 100 marks) both CIE and SEE marks put together.

The student is eligible to write Semester End Examination of the concerned subject, if the student scores $\geq 35\%$ (14 marks) of 40 Continuous Internal Examination (CIE) marks.

In case, the student appears for Semester End Examination (SEE) of the concerned subject but not scored minimum 35% of CIE marks (14 marks out of 40 internal marks), his performance in that subject in SEE shall stand cancelled inspite of appearing the SEE.

There is NO Computer Based Test (CBT) for R22 regulations.

The details of the end semester question paper pattern are as follows:

- 8.2.1** The semester end examinations (SEE), for theory subjects, will be conducted for 60 marks consisting of two parts viz. i) **Part- A** for 10 marks, ii) **Part - B** for 50 marks.
- Part-A is a compulsory question which consists of ten sub-questions from all units carrying equal marks.
 - Part-B consists of five questions (numbered from 2 to 6) carrying 10 marks each. Each of these questions is from each unit and may contain sub-questions. For each question there will be an “either” “or” choice, which means that there will be two questions from each unit and the student should answer either of the two questions.
 - The duration of Semester End Examination is 3 hours.
- 8.3** For practical subjects there shall be a Continuous Internal Evaluation (CIE) during the semester for 40 marks and 60 marks for semester end examination. Out of the 40 marks for internal evaluation:
1. A write-up on day-to-day experiment in the laboratory (in terms of aim, components/procedure, expected outcome) which shall be evaluated for 10 marks
 2. **10 marks for viva-voce** (or) tutorial (or) case study (or) application (or) poster presentation of the course concerned.
 3. Internal practical examination conducted by the laboratory teacher concerned shall be evaluated for 10 marks.
 4. The remaining 10 marks are for Laboratory Project, which consists of the Design (or) Software / Hardware Model Presentation (or) App Development (or) Prototype Presentation submission which shall be evaluated after completion of laboratory course and before semester end practical examination.

The Semester End Examination shall be conducted with an external examiner and the laboratory teacher. The external examiner shall be appointed from the other colleges which will be decided by the principal of the College.

In the Semester End Examination held for 3 hours, total 60 marks are divided and allocated as shown below:

1. 10 marks for write-up
 2. 15 for experiment/program
 3. 15 for evaluation of results
 4. 10 marks for presentation on another experiment/program in the same laboratory course and
 5. 10 marks for viva-voce on concerned laboratory course.
- The Student, in each subject, shall have to earn 35% of marks (i.e. 14 marks out of 40 marks) in CIE, 35% of marks (i.e. 21 marks out of 60) in SEE and Over all 40% of marks (i.e. 40 marks out of 100 marks) both CIE and SEE marks put together.

The student is eligible to write Semester End Examination of the concerned subject, if the student scores $\geq 35\%$ (14 marks) of 40 Continuous Internal Examination (CIE) marks.

In case, the student appears for Semester End Examination (SEE) of the concerned

subject but not scored minimum 35% of CIE marks (14 marks out of 40 internal marks), his performance in that subject in SEE shall stand cancelled inspite of appearing the SEE.

8.4 The evaluation of courses having ONLY internal marks in I Year I Semester and II Year II Semester is as follows:

1. I Year I, II Semester course (*ex., Elements of CE/ME/EEE/ECE/CSE/Seminar & etc*): The internal evaluation is for 50 marks and it shall take place during I Mid-Term examination and II Mid-Term examination. The average marks of two MID-Term examinations is the final for 50 marks. Student shall have to earn 40%, i.e 20 marks out of 50 marks from average of the two examinations. There shall be NO external evaluation. The student is deemed to have failed, if he (i) is absent as per schedule, or (ii) secures less than 40% marks in this course. Seminar evaluation conducted through day to day evaluation of the students in presentation/abstract.
2. II Year II Semester *Real-Time (or) Field-based Research Project* course: The internal evaluation is for 50 marks and it shall take place during I Mid-Term examination and II Mid-Term examination. The average marks of two Mid-Term examinations is the final for 50 marks. Student shall have to earn 40%, i.e 20 marks out of 50 marks from average of the two examinations. There shall be NO external evaluation. The student is deemed to have failed, if he (i) does not submit a report on the Project, or (ii) does not make a presentation of the same before the internal committee as per schedule, or (ii) secures less than 40% marks in this course.

8.5 There shall be an Industry training (or) Internship (or) Industry oriented Mini-project (or) Skill Development Courses (or) Paper presentation in reputed journal (or) Industry Oriented Mini Project in collaboration with an industry of their specialization. Students shall register for this immediately after II-Year II Semester Examinations and pursue it during summer vacation/semester break & during III Year without effecting regular course work. Internship at reputed organization (or) Skill development courses (or) Paper presentation in reputed journal (or) Industry Oriented Mini Project shall be submitted in a report form and presented before the committee in III-year II semester before end semester examination. It shall be evaluated for 100 external marks. The committee consists of an External Examiner, Head of the Department, Supervisor of the Industry Oriented Mini Project (or) Internship etc, Internal Supervisor and a Senior Faculty Member of the Department. There shall be **NO internal marks** for Industry Training (or) Internship (or) Mini-Project (or) Skill Development Courses (or) Paper Presentation in reputed journal (or) Industry Oriented Mini Project.

8.6 The UG project shall be initiated at the end of the IV Year I Semester and the duration of the project work is one semester. The student must present Project Stage – I during IV Year I Semester before II Mid examinations, in consultation with his Supervisor, the title, objective and plan of action of his Project work to the departmental committee for approval before commencement of IV Year II Semester. Only after obtaining the approval of the departmental committee, the student can start his project work.

8.7 UG project work shall be carried out in two stages: Project Stage – I for approval of project before Mid-II examinations in IV Year I Semester and Project Stage – II during IV Year II Semester. Student has to submit project work report at the end of IV Year II Semester. The project shall be evaluated for 100 marks before commencement of SEE Theory examinations.

8.8 For Project Stage – I, the departmental committee consisting of Head of the Department, project supervisor and a senior faculty member shall approve the project work to begin before II Mid-Term examination of IV Year I Semester. The student is deemed to be not eligible to register for the Project work, if he does not submit a report on Project Stage - I or does not make a presentation of the same before the evaluation committee as per schedule.

A student who has failed may reappear once for the above evaluation, when it is scheduled again; if he fails in such ‘one reappearance’ evaluation also, he has to reappear for the same in the next subsequent semester, as and when it is scheduled.

8.9 For Project Stage – II, the external examiner shall evaluate the project work for 60 marks and the internal project committee shall evaluate it for 40 marks. Out of 40 internal marks, the departmental committee consisting of Head of the Department, Project Supervisor and a Senior Faculty Member shall evaluate the project work for 20 marks and Project Supervisor shall evaluate for 20 marks. The topics for Industry Oriented Mini Project/ Internship/SDC etc. and the main Project shall be different from the topic already taken. The student is deemed to have failed, if he (i) does not submit a report on the Project, or (ii) does not make a presentation of the same before the External Examiner as per schedule, or (iii) secures less than 40% marks in the sum total of the CIE and SEE taken together.

For conducting viva-voce of project, Principal/COE selects an external examiner from the list of experts in the relevant branch submitted by the Principal of the College.

A student who has failed, may reappear once for the above evaluation, when it is scheduled again; if student fails in such ‘one reappearance’ evaluation also, he has to reappear for the same in the next subsequent semester, as and when it is scheduled.

8.10 A student shall be given only one time chance to re-register for a maximum of two subjects in a semester:

- If the internal marks secured by a student in the Continuous Internal Evaluation marks for 40 (Sum of average of two mid-term examinations consisting of Objective & descriptive parts, Average of two Assignments & Subject Viva- voce/PPT/ Poster presentation/ Case Study on a topic in the concerned subject) are less than 35% and failed in those subjects.

A student must re-register for the failed subject(s) for 40 marks within four weeks of commencement of the classwork in next academic year.

In the event of the student taking this chance, his Continuous Internal Evaluation marks for 40 and Semester End Examination marks for 60 obtained in the previous attempt stand cancelled.

8.11 For mandatory non-credit Audit courses, The internal evaluation is for 50 marks and it shall take place during I Mid-Term examination and II Mid-Term examination. The average marks of two MID-Term examinations is the final for 50 marks. Student shall have to earn 40%, i.e 20 marks out of 50 marks from average of the two examinations. There shall be NO external evaluation. The student is deemed to have failed, if he (i) is absent as per schedule, or (ii) secures less than 40% marks in this course.

8.12 For Mandatory Non-Credit Courses offered in a Semester, after securing $\geq 65\%$ attendance and has secured not less than 40% of marks in the sum Total. Then the student is **PASS** and will be qualified for the award of the degree. No marks or Letter Grade shall be allotted for these courses/activities. However, for non credit courses ‘**Satisfactory**’ or ‘**Unsatisfactory**’ shall be indicated instead of the letter grade and this will not be counted for the computation of SGPA/CGPA.

9.0 Grading Procedure

9.1 Grades will be awarded to indicate the performance of students in each Theory Subject, Laboratory/Practicals/ Industry-Oriented Mini Project/Internship/SDC and Project Stage. Based on the percentage of marks obtained (Continuous Internal Evaluation plus Semester End Examination, both taken together) as specified in item 8 above, a corresponding letter grade shall be given.

9.2 As a measure of the performance of a student, a 10-point absolute grading system using the following letter grades (as per UGC/AICTE guidelines) and corresponding percentage of marks shall be followed:

% of Marks Secured in a Subject/Course (Class Intervals)	Letter Grade (UGC Guidelines)	Grade Points
Greater than or equal to 90%	O (Outstanding)	10
80 and less than 90%	A ⁺ (Excellent)	9
70 and less than 80%	A (Very Good)	8
60 and less than 70%	B ⁺ (Good)	7
50 and less than 60%	B (Average)	6
40 and less than 50%	C (Pass)	5
Below 40%	F (FAIL)	0
Absent	Ab	0

9.3 A student who has obtained an ‘**F**’ grade in any subject shall be deemed to have ‘**failed**’ and is required to reappear as a ‘supplementary student’ in the semester end examination, as and when offered. In such cases, internal marks in those subjects will remain the same as those obtained earlier.

9.4 To a student who has not appeared for an examination in any subject, ‘**Ab**’ grade will be allocated in that subject, and he is deemed to have ‘**Failed**’. A student will be required to reappear as a ‘supplementary student’ in the semester end examination, as and when offered next. In this case also, the internal marks in those subjects will remain the same as those obtained earlier.

9.5 A letter grade does not indicate any specific percentage of marks secured by the student, but it indicates only the range of percentage of marks.

- 9.6 A student earns Grade Point (GP) in each subject/ course, on the basis of the letter grade secured in that subject/ course. The corresponding 'Credit Points' (CP) are computed by multiplying the grade point with credits for that particular subject/ course.

Credit Points (CP) = Grade Point (GP) x Credits For a course

- 9.7 A student passes the subject/ course only when **GP $\geq$ 5 ('C' grade or above)**
- 9.8 The Semester Grade Point Average (SGPA) is calculated by dividing the sum of credit points (Σ CP) secured from all subjects/ courses registered in a semester, by the total number of credits registered during that semester. SGPA is rounded off to **two** decimal places. SGPA is thus computed as

$$\text{SGPA} = \{ \sum_{i=1}^N C_i G_i \} / \{ \sum_{i=1}^N C_i \} \dots \text{For each Semester,}$$

where 'i' is the subject indicator index (considering all subjects in a semester), 'N' is the no. of subjects '**registered**' for the semester (as specifically required and listed under the course structure of the parent department), C_i is the no. of credits allotted to the i^{th} subject, and G_i represents the grade points (GP) corresponding to the letter grade awarded for that i^{th} subject.

- 9.9 The Cumulative Grade Point Average (CGPA) is a measure of the overall cumulative performance of a student in all semesters considered for registration. The CGPA is the ratio of the total credit points secured by a student in **all** registered courses (of 160) in **all** semesters, and the total number of credits registered in **all** the semesters. CGPA is rounded off to **two** decimal places. CGPA is thus computed from the I year II semester onwards at the end of each semester as per the formula

$$\text{CGPA} = \{ \sum_{j=1}^M C_j G_j \} / \{ \sum_{j=1}^M C_j \} \dots \text{for all S Semesters registered}$$

(i.e., up to and inclusive of S Semesters, $S \geq 2$),

where 'M' is the TOTAL no. of Subjects (as specifically required and listed under the Course Structure of the parent Department) the Student has 'REGISTERED' from the 1st Semester onwards up to and inclusive of the Semester S (obviously $M > N$), 'j' is the Subject indicator index (takes into account all Subjects from 1 to S Semesters), C_j is the no. of Credits allotted to the jth Subject, and G_j represents the Grade Points (GP) corresponding to the Letter Grade awarded for that jth Subject. After registration and completion of I Year I Semester however, the SGPA of that Semester itself may be taken as the CGPA, as there are no cumulative effects.

Illustration of calculation of SGPA:

Course/Subject	Credits	Letter Grade	Grade Points	Credit Points
Course 1	4	A	8	$4 \times 8 = 32$
Course 2	4	O	10	$4 \times 10 = 40$
Course 3	4	C	5	$4 \times 5 = 20$
Course 4	3	B	6	$3 \times 6 = 18$
Course 5	3	A+	9	$3 \times 9 = 27$
Course 6	3	C	5	$3 \times 5 = 15$
	21			152

$$\text{SGPA} = 152/21 = 7.24$$

Illustration of Calculation of CGPA up to 3rd Semester:

Semester	Course/ Subject Title	Credits Allotted	Letter Grade Secured	Corresponding Grade Point (GP)	Credit Points (CP)
I	Course 1	3	A	8	24
I	Course 2	3	O	10	30
I	Course 3	3	B	6	18
I	Course 4	4	A	8	32
I	Course 5	3	A+	9	27
I	Course 6	4	C	5	20
II	Course 7	4	B	6	24
II	Course 8	4	A	8	32
II	Course 9	3	C	5	15
II	Course 10	3	O	10	30
II	Course 11	3	B+	7	21
II	Course 12	4	B	6	24
II	Course 13	4	A	8	32
II	Course 14	3	O	10	30
III	Course 15	2	A	8	16
III	Course 16	1	C	5	5
III	Course 17	4	O	10	40
III	Course 18	3	B+	7	21
III	Course 19	4	B	6	24
III	Course 20	4	A	8	32
III	Course 21	3	B+	7	21
	Total Credits	69		Total Credit Points	518

$$\text{CGPA} = 518/69 = 7.51$$

The calculation process of CGPA illustrated above will be followed for each subsequent semester until 8th semester. The CGPA obtained at the end of 8th semester will become the final CGPA secured for entire B.Tech. programme.

- 9.10** For merit ranking or comparison purposes or any other listing, **only** the ‘**rounded off**’ values of the CGPAs will be used.
- 9.11** SGPA and CGPA of a semester will be mentioned in the semester Memorandum of Grades if all subjects of that semester are passed in first attempt. Otherwise the SGPA and CGPA shall be mentioned only on the Memorandum of Grades in which sitting he passed his last exam in that semester. However, mandatory courses will not be taken into consideration.

10.0 Passing Standards

- 10.1** A student shall be declared successful or 'passed' in a semester, if he secures a GP ≥ 5 ('C' grade or above) in every subject/course in that semester (i.e. when the student gets an SGPA ≥ 5.0 at the end of that particular semester); and he shall be declared successful or 'passed' in the entire undergraduate programme, only when gets a CGPA ≥ 5.00 ('C' grade or above) for the award of the degree as required.
- 10.2** After the completion of each semester, a grade card or grade sheet shall be issued to all the registered students of that semester, indicating the letter grades and credits earned. It will show the details of the courses registered (course code, title, no. of credits, grade earned, etc.) and credits earned. **There is NO exemption of credits in any case.**

11.0 Declaration of results

- 11.1** Computation of SGPA and CGPA are done using the procedure listed in 9.6 to 9.9.
- 11.2** For final percentage of marks equivalent to the computed final CGPA, the following formula may be used.
- $$\% \text{ of Marks} = (\text{final CGPA} - 0.5) \times 10$$

12.0 Award of Degree

- 12.1** A student who registers for all the specified subjects/ courses as listed in the course structure and secures the required number of 160 credits (with CGPA ≥ 5.0), within 8 academic years from the date of commencement of the first academic year, shall be declared to have '**qualified**' for the award of B.Tech. degree in the branch of Engineering selected at the time of admission.
- 12.2** A student who qualifies for the award of the degree as listed in item 12.1 shall be placed in the following classes.
- 12.3** A student with final CGPA (at the end of the undergraduate programme) > 8.00 , and fulfilling the following conditions - shall be placed in '**First Class with Distinction**'. However, he
- (i) Should have passed all the subjects/courses in '**First Appearance**' within the first 4 academic years (or 8 sequential semesters) from the date of commencement of first year first semester.
 - (ii) Should not have been detained or prevented from writing the semester end examinations in any semester due to shortage of attendance or any other reason.
- A student not fulfilling any of the above conditions with final CGPA > 8 shall be placed in '**First Class**'.
- 12.4** Students with final CGPA (at the end of the undergraduate programme) ≥ 7.0 but < 8.00 shall be placed in '**First Class**'.
- 12.5** Students with final CGPA (at the end of the undergraduate programme) ≥ 6.00 but < 7.00 , shall be placed in '**Second Class**'.
- 12.6** All other students who qualify for the award of the degree (as per item 12.1), with final CGPA (at the end of the undergraduate programme) ≥ 5.00 but < 6 , shall be placed in '**pass class**'.

12.7 A student with final CGPA (at the end of the undergraduate programme) < 5.00 will not be eligible for the award of the degree.

12.8 Students fulfilling the conditions listed under item 12.3 alone will be eligible for award of '**Gold Medal**'.

12.9 Award of 2-Year B.Tech. Diploma Certificate

1. A student is awarded 2-Year UG Diploma Certificate in the concerned engineering branch on completion of all the academic requirements and earned all the 80 credits (with in 4 years from the date of admission) upto B. Tech. – II Year – II Semester, if the student want to exit the 4-Year B. Tech. program. The student **once opted and awarded for 2-Year UG Diploma Certificate, the student will not be permitted to join** in B. Tech. III Year – I Semester and continue for completion of remaining years of study for 4-Year B. Tech. Degree.
2. A student may be permitted to take one year break after completion of II Year – II Semester or B. Tech. – III Year – II Semester (with Principal permission through the HOD of the department well in advance) and can re-enter the course in **next Academic Year in the same college** and complete the course on fulfilling all the academic credentials within a stipulated duration i.e. double the duration of the course (Ex. within 8 Years for 4-Year program).

13.0 Withholding of results

13.1 If the student has not paid the fees to the College at any stage, or has dues pending due to any reason whatsoever, or if any case of indiscipline is pending, the result of the student may be withheld, and the student will not be allowed to go into the next higher semester. The award or issue of the degree may also be withheld in such cases.

14.0 Transitory Regulations

A. For students detained due to shortage of attendance:

1. A Student who has been detained in I year of R18 Regulations due to lack of attendance, shall be permitted to join I year I Semester of R22 Regulations and he is required to complete the study of B.Tech./B. Pharmacy programme within the stipulated period of eight academic years from the date of first admission in I Year.
2. A student who has been detained in any semester of II, III and IV years of R18 regulations for want of attendance, shall be permitted to join the corresponding semester of R22 Regulations and is required to complete the study of B.Tech./B. Pharmacy within the stipulated period of eight academic years from the date of first admission in I Year. The R22 Academic Regulations under which a student has been readmitted shall be applicable to that student from that semester. See rule (C) for further Transitory Regulations.
3. For students detained due to shortage of credits: A student of MLR18 Regulations who has been detained due to lack of credits, shall be promoted to the next semester of R22 Regulations only after acquiring the required number of credits as per the corresponding regulations of his/her first admission. The total credits required are 160 including both MLR18 & R22 regulations. The student is required to complete the study of B.Tech. within the stipulated period of eight academic years from the year of first admission. The R22 Academic Regulations are applicable to a student from the year of readmission. See rule (C) for further Transitory Regulations.

B. For readmitted students in R22 Regulations:

4. A student who has failed in any subject under any regulation has to pass those subjects in the same regulations.
5. The maximum credits that a student acquires for the award of degree, shall be the sum of the total number of credits secured in all the regulations of his/her study including R22 Regulations. **There is NO exemption of credits in any case.**
6. If a student is readmitted to R22 Regulations and has any subject with 80% of syllabus common with his/her previous regulations, that particular subject in R22 Regulations will be substituted by another subject to be suggested by the Concerned BOS.

15.0 Student Transfers

- 15.1 There shall be no branch transfers after the completion of admission process.
- 15.2 There shall be no transfers from one college/stream to another within the constituent colleges and units of Jawaharlal Nehru Technological University Hyderabad.
- 15.3 The students seeking transfer to colleges affiliated to JNTUH from various other Universities/institutions have to pass the failed subjects which are equivalent to the subjects of JNTUH, and also pass the subjects of JNTUH which the students have not studied at the earlier institution. Further, though the students have passed some of the subjects at the earlier institutions, if the same subjects are prescribed in different semesters of JNTUH, the students have to study those subjects in JNTUH in spite of the fact that those subjects are repeated.
- 15.4 The transferred students from other Universities/Institutions to College who are on rolls are to be provided one chance to write the CBT (for internal marks) in the **equivalent subject(s)** as per the clearance letter issued by the University.
- 15.5 The college have to provide one chance to write the internal examinations in the **equivalent subject(s)** to the students transferred from other universities/institutions to JNTUH autonomous affiliated colleges who are on rolls, as per the clearance (equivalence) letter issued by the University.

16.0 Malpractice Prevention Committee

A malpractice prevention committee shall be constituted to examine and punish the students who involve in malpractice / indiscipline in examinations. The committee shall consist of:

- a) Controller of examinations - Chairman
- b) Addl. Controller of examinations.- Member Convener
- c) Subject expert - member
- d) Head of the department of which the student belongs to. - Member
- e) The invigilator concerned - member

The committee shall conduct the meeting after taking explanation of the student and punishment will be awarded by following the malpractice rules meticulously.

Any action on the part of candidate at the examination like trying to get undue advantage in the performance at examinations or trying to help another, or derive the same through unfair means is punishable according to the provisions contained hereunder. The involvement of the Staff who are in charge of conducting examinations, valuing examination papers and preparing / keeping records of documents relating to the examinations, in such acts (inclusive of providing incorrect or misleading information)

that infringe upon the course of natural justice to one and all concerned at the examination shall be viewed seriously and will be recommended for appropriate punishment after thorough enquiry.

17.0 Scope

- 17.1 The academic regulations should be read as a whole, for the purpose of any interpretation.
- 17.2 In case of any doubt or ambiguity in the interpretation of the above rules, the decision of the Chairman of Academic Council is final.
- 17.3 The College may change or amend the academic regulations, course structure or syllabi at any time, and the changes or amendments made shall be applicable to all students with effect from the dates notified by the college authorities.
- 17.4 Where the words “he”, “him”, “his”, occur in the regulations, they include “she”, “her”, “hers”.

ACADEMIC REGULATIONS FOR B.TECH (LATERAL ENTRY SCHEME)
FROM THE AY 2023-24

1. Eligibility for the award of B.Tech Degree (LES)

The LES students after securing admission shall pursue a course of study for not less than three academic years and not more than six academic years.

2. The student shall register for 120 credits and secure 120 credits with CGPA ≥ 5 from II year to IV-year B.Tech. programme (LES) for the award of B.Tech. degree.
3. The students, who fail to fulfil the requirement for the award of the degree in six academic years from the year of admission, shall forfeit their seat in B.Tech.
4. The attendance requirements of B. Tech. (Regular) shall be applicable to B.Tech. (LES).
5. **Promotion rule**

S. No	Promotion	Conditions to be fulfilled
1	Second year first semester to second year second semester	Regular course of study of second year first semester.
2	Second year second semester to third year first semester	(i) Regular course of study of second year second semester. (ii) Must have secured at least 24 credits out of 40 credits i.e., 60% credits up to second year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.
3	Third year first semester to third year second semester	Regular course of study of third year first semester.
4	Third year second semester to fourth year first semester	i) Regular course of study of third year second semester. ii) Must have secured at least 48 credits out of 80 credits i.e., 60% credits up to third year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.
5	Fourth year first semester to fourth year second semester	Regular course of study of fourth year first semester.

6. All the other regulations as applicable to B. Tech. 4-year degree course (Regular) will hold good for B. Tech. (Lateral Entry Scheme).

7. LES students are not eligible for 2-Year B. Tech. Diploma Certificate.

Malpractices Rules

Disciplinary Action For / Improper Conduct in Examinations

	Nature of Malpractices/ Improper conduct	Punishment
	If the student:	
1. (a)	Possesses or keeps accessible in examination hall, any paper, note book, programmable calculators, cell phones, pager, palm computers or any other form of material concerned with or related to the subject of the examination (theory or practical) in which student is appearing but has not made use of (material shall include any marks on the body of the student which can be used as an aid in the subject of the examination).	Expulsion from the examination hall and cancellation of the performance in that subject only.
(b)	Gives assistance or guidance or receives it from any other student orally or by any other body language methods or communicates through cell phones with any student or persons in or outside the exam hall in respect of any matter.	Expulsion from the examination hall and cancellation of the performance in that subject only of all the students involved. In case of an outsider, he will be handed over to the police and a case is registered against him.
2.	Has copied in the examination hall from any paper, book, programmable calculators, palm computers or any other form of material relevant to the subject of the examination (theory or practical) in which the student is appearing.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the student has already appeared including practical examinations and project work and shall not be permitted to appear for the remaining examinations of the subjects of that semester/year. The hall ticket of the student is to be cancelled
3.	Impersonates any other student in connection with the examination.	The student who has impersonated shall be expelled from examination hall. The student is also debarred and forfeits the seat. The performance of the original student who has been impersonated, shall be cancelled in all the subjects of the examination (including practicals and project work) already appeared and shall not be allowed to appear for examinations of the remaining subjects of that semester/year. The student is also debarred for two consecutive semesters from class work and all college examinations. The continuation of the course by the student is subject to the academic regulations in connection with forfeiture of seat. If the imposter is an outsider, he will be handed over to the police and a case is registered against him.

4.	Smuggles in the answer book or additional sheet or takes out or arranges to send out the question paper during the examination or answer book or additional sheet, during or after the examination.	Expulsion from the examination hall and cancellation of performance in that subject and all the other subjects the student has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The student is also debarred for two consecutive semesters from class work and all college examinations. The continuation of the course by the student is subject to the academic regulations in connection with forfeiture of seat.
5.	Uses objectionable, abusive or offensive language in the answer paper or in letters to the examiners or writes to the examiner requesting him to award pass marks.	Cancellation of the performance in that subject.
6.	Refuses to obey the orders of the chief superintendent/COE/ACoE/any officer on duty or misbehaves or creates disturbance of any kind in and around the examination hall or organizes a walk out or instigates others to walk out, or threatens the officer or any person on duty in or outside the examination hall of any injury to his person or to any of his relations whether by words, either spoken or written or by signs or by visible representation, assaults the officer-in-charge, or any person on duty in or outside the examination hall or any of his relations, or indulges in any other act of misconduct or mischief which result in damage to or destruction of property in the examination hall or any part of the college campus or engages in any other act which in the opinion of the officer on duty amounts to use of unfair means or misconduct or has the tendency to disrupt the orderly conduct of the examination.	In case of students of the college, they shall be expelled from examination halls and cancellation of their performance in that subject and all other subjects the student(s) has (have) already appeared and shall not be permitted to appear for the remaining examinations of the subjects of that semester/year. The students also are debarred and forfeit their seats. In case of outsiders, they will be handed over to the police and a police case is registered against them.
7.	Leaves the exam hall taking away answer script or intentionally tears off the script or any part thereof inside or outside the examination hall.	Expulsion from the examination hall and cancellation of performance in that subject and all the other subjects the student has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The student is also debarred for two consecutive semesters from class work and all college examinations. The continuation of the course by the student is subject to the academic regulations in connection with forfeiture of seat.

8.	Possesses any lethal weapon or firearm in the examination hall.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the student has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The student is also debarred and forfeits the seat.
9.	If student of the college, who is not a student for the particular examination or any person not connected with the college indulges in any malpractice or improper conduct mentioned in clause 6 to 8.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the student has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The student is also debarred and forfeits the seat. Person(s) who do not belong to the college will be handed over to the police and, a police case will be registered against them.
10.	Comes in a drunken condition to the examination hall.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the student has already appeared for including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year.
11.	Copying detected on the basis of internal evidence, such as, during valuation or during special scrutiny.	Cancellation of the performance in that subject and all other subjects the student has appeared for including practical examinations and project work of that semester/year examinations.
12	If any malpractice is detected which is not covered in the above clauses 1 to 11 shall be reported to the Principal for further action to award a suitable punishment.	

Malpractices identified by squad or special invigilators

1. Punishments to the students as per the above guidelines.
2. Punishment for staff: (if the squad reports that the staff is also involved in encouraging malpractices)
 - a. A show-cause notice shall be issued to the staff.
 - b. Impose a suitable fine on the staff.

* * * * *

COURSE STRUCTURE

COURSE STRUCTURE

B. TECH – Aeronautical Engineering

REGULATIONS: R22

I YEAR I SEMESTER									
Induction program for one weeks									
Code	Course	Category	Periods per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		CIE	SEE	Total
A6BS01	Linear Algebra and Calculus	BSC	3	1	0	4	40	60	100
A6BS07	Applied Physics	BSC	3	1	0	4	40	60	100
A6HS01	English for Skill Enhancement	HSMC	3	0	0	3	40	60	100
A6ME01	Engineering Graphics	ESC	1	0	4	3	40	60	100
A6BS08	Applied Physics Lab	BSC	0	0	3	1.5	40	60	100
A6HS02	English Language and Communication Skills Lab	HSMC	0	0	3	1.5	40	60	100
A6ME03	Engineering workshop Practices	ESC	0	1	2	2	40	60	100
A6HS04	Seminar	HSMC	0	0	2	1	50	--	50
TOTAL			13	3	14	20	330	420	750
I YEAR II SEMESTER									
Code	Course	Category	Periods per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		CIE	SEE	Total
A6BS02	Numerical Methods and Integral Transforms	BSC	3	1	0	4	40	60	100
A6BS09	Engineering Chemistry	BSC	3	1	0	4	40	60	100
A6ME05	Engineering Mechanics	ESC	4	0	0	4	40	60	100
A6CS02	Programming for Problem Solving	ESC	3	0	0	3	40	60	100
A6AE01	Elements of Aeronautical Engineering	ESC	0	0	3	2	40	60	100
A6BS10	Engineering Chemistry Lab	BSC	1	0	2	1.5	40	60	100
A6CS03	Programming for Problem Solving Lab	ESC	0	0	3	1.5	40	60	100
A6BS11	Environmental Science	MC	2	0	0	0	50	--	50
TOTAL			16	2	8	20	320	420	750

B. Tech.- III-Semester									
Course Code	Course Title	Category	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		CIE	SEE	Total
A6AE02	Airplane Performance	PCC	3	-	-	3	40	60	100
A6AE03	Solid Mechanics for Aeronautics	PCC	3	-	-	3	40	60	100
A6AE04	Aerodynamics	PCC	3	-	-	3	40	60	100
A6AE05	Aero Thermodynamics	ESC	3	1	-	4	40	60	100
A6EE60	Basic Electrical and Electronics Engineering	ESC	3	-	-	3	40	60	100
A6AE06	Solid Mechanics for Aeronautics Lab	ESC	-	-	3	1.5	40	60	100
A6EE61	Basic Electrical and Electronics Engineering Lab	ESC	-	-	2	1	40	60	100
A6AE07	Aerodynamics Lab	PCC	-	-	3	1.5	40	60	100
TOTAL			15	1	8	20	320	480	800
Mandatory Course (Non-Credit)									
A6HS13	Intellectual Property Rights	HSMC	2	-	-	-	40	60	100
B. Tech.- IV-Semester									
Course Code	Course Title	Category	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		CIE	SEE	Total
A6AE08	IoT Based Drone Systems	ESC	3	-	-	3	40	60	100
A6AE09	Aerospace Propulsion	PCC	3	-	-	3	40	60	100
A6AE10	Aerospace Structures	PCC	3	-	-	3	40	60	100
A6BS05	Probability, Statistics and Complex Analysis	BSC	3	1	-	4	40	60	100
A6AE11	Flight Simulation and Propulsion Lab	PCC	-	-	3	1.5	40	60	100
A6AE61	IOT Lab	ESC	-	-	2	1	40	60	100
A6AE12	Aircraft Interior Design Lab	PCC	-	-	3	1.5	40	60	100
A6AE13	Python Computing Lab	ESC	-	-	2	1	40	60	100
A6AE14	Real-Time /Field-Based Project	PWC	-	-	4	2	40	60	100
TOTAL			12	1	14	20	360	540	900
Mandatory Course (Non-Credit)									

A6HS05	Gender Sensitization	HSMC	2	-	-	-	40	60	100
B. Tech-V-Semester									
Course Code	Course Title	Category	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		CIE	SEE	Total
A6AE15	Smart Aerospace Structures	PCC	3	1	-	4	40	60	100
A6AE16	AI for Aerodynamics	PCC	3	1	-	4	40	60	100
A6AE17	Aircraft Stability and Control	PCC	3	-	-	3	40	60	100
A6AE18	Aircraft Production Technology	ESC	3	-	-	3	40	60	100
	OPEN ELECTIVE - I	OEC	3	-	-	3	40	60	100
A6AE21	Aerospace Structures Lab	PCC	-	-	3	1.5	40	60	100
A6AE22	Aircraft Production Technology Lab	ESC	-	-	3	1.5	40	60	100
TOTAL			15	2	6	20	280	420	700
Mandatory Course (Non-Credit)									
A6HS10	Human Values and Professional Ethics	HSMC	2	-	-	-	40	60	100

B. Tech.- VI-Semester									
Course Code	Course Title	Category	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		CIE	SEE	Total
A6AE23	UAV Design	PCC	3	-	-	3	40	60	100
A6AE24	Air Transportation System	PCC	3	-	-	3	40	60	100
	OPEN ELECTIVE-II	OEC	3	-	-	3	40	60	100
	PROFESSIONAL ELECTIVE - I	PEC	3	-	-	3	40	60	100
	PROFESSIONALELECTIVE - II	PEC	3	-	-	3	40	60	100
A6AE35	UAV Design Lab	PCC	-	-	3	1.5	40	60	100
A6AE36	Computational Structural Analysis Lab	PCC	-	-	3	1.5	40	60	100
A6AE37	Industry oriented Mini project/ internship	PWC	-	-	4	2	40	60	100
TOTAL			15	-	10	20	320	480	800

B. Tech.-VII-Semester									
Course Code	Course Title	Category	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		CIE	SEE	Total
A6AE38	Composite Materials	PCC	3	-	-	3	40	60	100
A6AE39	Computational Fluid Dynamics	PCC	3	-	-	3	40	60	100
	OPEN ELECTIVE-III	OEC	3	-	-	3	40	60	100
	PROFESSIONAL ELECTIVE - III	PEC	3	-	-	3	40	60	100
	PROFESSIONAL ELECTIVE - IV	PEC	3	-	-	3	40	60	100
A6AE50	Composites and Analysis Lab	PCC	-	-	3	1.5	40	60	100
A6AE51	Computational Fluid Dynamics Lab	PCC	-	-	3	1.5	40	60	100
A6AE52	Research Project Phase-I	PWC	-	-	4	2	100	--	100
TOTAL			15	-	10	20	380	420	800

B.Tech.- VIII-Semester									
Course Code	Course Title	Category	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		CIE	SEE	Total
A6HS12	Management Science	HSMC	3	-	-	3	40	60	100
	PROFESSIONAL ELECTIVE – V	PEC	3	-	-	3	40	60	100
	PROFESSIONAL ELECTIVE– VI	PEC	3	-	-	3	40	60	100
A6AE61	Research Project Phase-II	PWC	-	-	22	11	100	100	200
TOTAL			9	-	22	20	220	280	500

PROFESSIONAL ELECTIVES			
PE-I		PE-II	
A6AE27	Vibration and Structural Dynamics	A6AE31	Computer Integrated Manufacturing
A6AE28	Experimental Stress Analysis	A6AE32	Aero Engine Repair and Maintenance
A6AE29	Fatigue and Fracture Mechanics	A6AE33	Mechanisms and Machine Design
A6AE30	Finite Element Analysis	A6AE34	Geometric Dimensions and Tolerance
PE-III		PE-IV	
A6AE42	Helicopter Engineering	A6AE46	Deep Learning in Flight Vehicle
A6AE43	Aero Elasticity	A6AE47	Industry 4.0
A6AE44	Avionics	A6AE48	AI&ML for UAV
A6AE45	Industrial Aerodynamics	A6AE49	3D Printing
PE-V		PE-VI	
A6AE53	Fundamentals of Combustion	A6AE57	Flight Scheduling and Operations
A6AE54	Rocket and Missiles	A6AE58	Aircraft systems and Instruments
A6AE55	Space Mechanics	A6AE59	Airframe Maintenance and Repair
A6AE56	Advanced Propulsion	A6AE60	Air Traffic Control and Aerodrome Design

OPEN ELECTIVE COURSES

OPEN ELECTIVE COURSE-I			
S. No.	Course Code	Course Name	Offering Department
1.	A6AE19	Fundamentals of Avionics	Aeronautical Engineering
2.	A6AE20	Introduction to Aerospace Technology	
OPEN ELECTIVE COURSE-II			
S. No.	Course Code	Course Name	Offering Department
1.	A6AE25	Introduction to Jets and Rockets	Aeronautical Engineering
2.	A6AE26	Non-Destructive Testing Methods	

OPEN ELECTIVE COURSE-III			
S. No.	Course Code	Course Name	Offering Department
1.	A6AE40	Introduction to Aircraft Industry	Aeronautical Engineering
2.	A6AE41	Unmanned Aerial Vehicles	

B. TECH I SEMESTER SYLLABUS

LINEAR ALGEBRA & CALCULUS

I Semester: Common for All (ECE/EEE/MECH/AERO/ CSE/IT/AI&ML/CS/DS)								
Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A6BS01	BSC	L	T	P	C	CIA	SEE	Total
		3	1	-	4	40	60	100
UNIT-I	MATRICES AND THEIR APPLICATIONS							
Real matrices: Symmetric-skew-symmetric and orthogonal matrices –Complex matrices: Hermitian, Skew – Hermitian and Unitary matrices –Elementary row and column transformations –Elementary matrix-Finding rank of a matrix by reducing to Echelon form and Normal form-Finding the inverse of a matrix using elementary row/column transformations (Gauss-Jordan method)-Consistency of system of linear equations (homogeneous and non-homogeneous) using the rank of a matrix –Solving m n and n n linear system of equations by Gauss Elimination, Gauss sie del Method								
UNIT-II	EIGEN VALUES, EIGEN VECTORS							
Eigen values and Eigen vectors and its properties (without proof), Cayley-Hamilton theorem (Statement and verification)-Finding inverse and powers of a matrix by Cayley-Hamilton theorem, Diagonalization of matrices. Quadratic forms and Nature of the Quadratic Forms, Reduction of Quadratic form to Canonical forms by Orthogonal Transformation.								
UNIT-III	ORDINARY DIFFERENTIAL EQUATIONS AND THEIR APPLICATIONS							
Introduction- Exact and reducible to Exact differential equations-Newton’s Law of cooling-Law of Growth and Decay. Linear differential equations of second and higher order with constant coefficients - Non-Homogeneous term of the type $Q(x) = e^{ax}$, $\sin ax$, $\cos ax$, $e^{ax}v(x)$, $x^n v(x)$ - Method of variation of parameters L-C-R Circuits.								
UNIT-IV	MULTIPLE INTEGRALS, BETA AND GAMMA FUNCTIONS							
Double and triple integrals (Cartesian and polar), Change of order of integration in double integrals, Change of variables (Cartesian to polar) in double integrals. Finding the area and volume of a region using double and triple integral. Beta- Gamma Functions and their Properties-Relation between them- Evaluation of improper integrals using Gamma and Beta functions.								
UNIT-V	CALCULUS OF SEVERAL VARIABLES							
Limit, Continuity - Partial derivative- Partial derivatives of higher order -Total derivative - Chain rule, Jacobians- functional dependence & independence. Applications: Maxima and Minima of functions of two variables without constraints and Lagrange’s method (with constraints).								
Text Books:								
1. Ervin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006. 2. B.S.Grewal, Higher Engineering Mathematics, Khanna publishers, 36th Edition, 2010.								
Reference Books:								
1. G.B.Thomas, calculus and analytical geometry,9th Edition, Pearson Reprint 2006. 2. N.P Bali and Manish Goyal ,A Text of Engineering Mathematics,Laxmi publications,2008. 3. E.L.Ince, Ordinary differential Equations.Dover publications.1958.								

APPLIED PHYSICS

I Semester: CSE, IT, CSE-Data Science, CSE-Cyber Security, CSE-AIML, CSIT, ECE, MECH, AERO, EEE										
Course Code:		Category		Hours / Week		Credits	Maximum Marks			
A6BS08		BSC		L	T	P	C	CIA	SEE	Total
				3	1	0	4	40	60	100
UNIT-I		Quantum Physics								
Black body radiation, Stefan-Boltzmann's law, Planck's radiation law (Qualitative treatment), Photoelectric effect Waves and Particles, de Broglie Hypothesis, Matter Waves, Davisson and Germer's experiment, Heisenberg's Uncertainty Principle, Schrodinger's Time Independent Wave Equation, Physical Significance of the wave Function, Particle in One Dimensional Potential Box.										
UNIT-II		Electronic properties of Materials & Band theory of solids								
Electronic properties of Materials: Classical free electron theory and Quantum free electron theories of metals, success and drawbacks, Bloch theorem, Kronig-Penny model (Qualitative treatment), E-k diagram, effective mass of electron. Band theory of solids: Origin of energy band formation in solids, Fermi energy level, Fermi-Dirac distribution law, Classification of materials as conductors, insulators and semiconductors.										
UNIT-III		Semi-conductors & Semiconductor Devices								
Semiconductors: Intrinsic and Extrinsic Semiconductors, formation of PN junction diode and its V-I characteristics, Direct and Indirect band gap semiconductors, Hall effect and its applications. Semiconductor Devices: Construction, working and V-I characteristics of PIN Diode, LED, Solar cell and their applications.										
UNIT-IV		Laser & Fiber Optics								
Laser: Characteristics of Laser, Absorption, Spontaneous and Stimulated emission of radiations. Lasing actions-Pumping mechanism, Meta stable state and Population inversion, Nd-YAG laser, CO2 laser, Applications of lasers in different fields. Fiber Optics: Structure of fibers, Total Internal Reflection, Acceptance angle – Numerical Aperture, Types of fibers- SI and GI fibers, Single and Multimode fibers - SMSI, MMSI, MMGI, Fiber Optic Communication system, Signal Degradation - Attenuation mechanism, Dispersion, Applications of fibers in different fields.										
UNIT-V		Physics of Quantum computing & Quantum gates								
Physics of Quantum computing: Idea of classical bits and qubits, advantages with qubits over classical bits, Bloch vector representation of state of qubits. Quantum gates- Single qubit logic gates- Pauli X, Y, Z and Hadmard gate in matrix form, Two level gates-CNOT and SWAP gates and representation in matrix form, Comments on No cloning theorem, Entanglement, Quantum Teleportation – Basic Idea, Quantum Key distribution protocol - BB84 protocol.										
Text Books:										
1. P.K Palanisamy, Engineering Physics, Sitech Publications, 2013, IVth Ed. 2. Nielsen M. A., I. L Chung, Quantum Computation & Quantum Information, Cambridge Univ. Press. 3. M. N. Avadhanulu, P.G. Kshirsagar & TVS Arun Murthy" A Text book of Engineering Physics"- S. Chand Publications, 11th Edition 2019.										
Reference Books:										
1. Quantum Physics, H.C. Verma, TBS Publication, 2 nd Edition 2012 2. Fundamentals of Physics – Halliday, Resnick and Walker, John Wiley & Sons, 11 th Edition, 2018. 3. Engineering Physics by Shatendra Sharma and Jyotsna Sharma, Pearson Publication, 2019 4. B.K Pandey and Chaturvedi, Engineering Physics, Cengage Learning, 2 nd Edition, 2022										

ENGLISH FOR SKILL ENHANCEMENT

I Semester: Common to All Branches								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6HS01	HSMC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
UNIT-I	‘Toasted English’ by R. K. Narayan							
Vocabulary	: The Concept of Word Formation -The Use of Prefixes and Suffixes - Acquaintance with Prefixes and Suffixes from Foreign Languages to form Derivatives - Synonyms and Antonyms							
Grammar	: Identifying Common Errors in Writing with Reference to Articles and Prepositions							
Reading:	: Reading and Its Importance- Techniques for Effective Reading							
Writing	: Sentence Structures – Use of Phrases and Clauses in Sentences – Importance of Proper Punctuation – Techniques for Writing precisely – Paragraph Writing – Types, Structures and Features of a Paragraph – Creating Coherence – Organizing Principles of Paragraphs in Documents.							
UNIT-II	‘Appro JRD’ by Sudha Murthy							
Vocabulary	: Words Often Misspelt – Homophones, Homonyms and Homographs							
Grammar	: Identifying Common Errors in Writing with Reference to Noun-pronoun Agreement and Subject-verb Agreement							
Reading	: Sub-Skills of Reading – Skimming and Scanning – Exercises for Practice							
Writing	: Nature and Style of Writing- Defining /Describing People, Objects, Places and Events – Classifying- Providing Examples or Evidence.							
UNIT-III	‘Lessons from Online Learning’ by F.Haider Alvi, Deborah Hurst et al							
Vocabulary	: Words Often Confused – Words from Foreign Languages and their Use in English.							

Grammar	: Identifying Common Errors in Writing with Reference to Misplaced Modifiers and Tenses
Reading	: Sub-Skills of Reading – Intensive Reading and Extensive Reading – Exercises for Practice
Writing	: Format of a Formal Letter-Writing Formal Letters E.g., Letter of Complaint, Letter of Requisition, Email Etiquette, Job Application with CV/Resume
UNIT-IV	Art and Literature
Vocabulary	: Standard Abbreviations in English
Grammar	: Redundancies and Clichés in Oral and Written Communication
Reading	: Survey, Question, Read, Recite and Review (SQ3R Method) - Exercises for Practice
Writing	: Writing Practices- Essay Writing-Writing Introduction and Conclusion -Précis Writing
UNIT-V	Go, Kiss the World' by Subroto Bagchi
Vocabulary	: Technical Vocabulary and their Usage
Grammar	: Common Errors in English (<i>Covering all the other aspects of grammar which were not covered in the previous units</i>)
Reading	: Reading Comprehension-Exercises for Practice
Writing	: Technical Reports- Introduction – Characteristics of a Report – Categories of Reports Formats- Structure of Reports (Manuscript Format) -Types of Reports - Writing a Report
Text Books:	
1. "English: Language, Context and Culture" by Orient BlackSwan Pvt. Ltd, Hyderabad.2022. Print.	
Reference Books:	
1. Effective Academic Writing by Liss and Davis (OUP) 2. Richards, Jack C. (2022) Interchange Series. Introduction, 1,2,3. Cambridge University Press 3. Wood, F.T. (2007). Remedial English Grammar. Macmillan. 4. Chaudhuri, Santanu Sinha. (2018). Learn English: A Fun Book of Functional Language, Grammar and Vocabulary. (2 nd ed.,). Sage Publications India Pvt. Ltd. 5. (2019). Technical Communication. Wiley India Pvt. Ltd. 6. Vishwamohan, Aysha. (2013). English for Technical Communication for Engineering Students. Mc Graw-Hill Education India Pvt. Ltd. 7. Swan, Michael. (2016). Practical English Usage. Oxford University Press. Fourth Edition	

ENGINEERING GRAPHICS

I Semester (AERO, EEE & MECH)								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6ME01	ESC	L	T	P	C	CIE	SEE	Total
		1	-	4	3	40	60	100
UNIT-I	INTRODUCTION TO ENGINEERING GRAPHICS							
Introduction to Engineering Graphics: Principles and their significance, Introduction to Computer Aided Drafting: Initial Setup Commands, Draw Commands, Modify Commands, 2D Drawings-Simple Exercises.								
UNIT-II	ENGINEERING CURVES							
Engineering Curves: Ellipse, Parabola, and Hyperbola (General Method only). Special curves: Cycloids, Involutes.								
UNIT-III	ORTHOGRAPHIC PROJECTIONS OF POINTS, LINES AND PLANES							
Principles of Orthographic Projections – conventions – First and Third angle projections. Projections of points, Projection of lines, Lines inclined to both the planes. Projections of Planes: Projections of regular planes inclined to both planes.								
UNIT-IV	PROJECTIONS OF SOLIDS AND DEVELOPMENT OF SURFACES							
Projection of Solids: Regular Solids inclined to both planes Development of Surfaces: Theory of development, development of lateral surface with base.								
UNIT-V	ISOMETRIC VIEW AND ORTHOGRAPHIC VIEWS - CONVERSION							
Divisions of pictorial projection, theory of Isometric Drawing- Isometric view and Isometric projections. Drawing Isometric circles, Dimensioning Isometric objects. Conversion of Isometric view to Orthographic views and Orthographic views to isometric view.								
Text Books:								
1. Bhatt N.D., Panchal V.M. & Ingle P.R., Engineering Drawing, Charotar Publishing House. 2. Agrawal B. & Agrawal C. M., Engineering Graphics, TMH Publication.								
Reference Books:								
1. D.M. Kulkarni, A.P.Rastogi, A.K. Sarka “Engineering Graphics with AutoCAD” PHI publications 2. Narayana, K.L. & P Kannaiah, Text book on Engineering Drawing, Scitech Publishers. 3. Shah, M.B. & Rana B.C. , Engineering Drawing and Computer Graphics, Pearson Education.								

APPLIED PHYSICS LAB

I Semester: CSE, IT, CSE-Data Science, CSE-Cyber Security, CSE-AIML, CSIT, ECE, MECH, AERO, EEE								
Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A6BS08	BSC	L	T	P	C	CIA	SEE	Total
		0	0	3	1.5	40	60	100
LIST OF EXPERIMENTS								
Experiment-1	ENERGY GAP OF P-N JUNCION DIODE: To determine the energy gap of a given semiconductor diode							
Experiment-2	SOLAR CELL: To study the V-I and V-P characteristics and determine the fill factor of solar cell							
Experiment-3	LIGHT EMITTING DIODE: To study the characteristics of LED by plotting V-I graph and determine the threshold value of given LEDs							
Experiment-4	HALL EFFECT: To determine Hall co-efficient and charge carrier concentration of a given semiconductor							
Experiment-5	PIN PHOTO DIODE: To study the V-I Characteristics of Photo Diode with respect to intensity of light							
Experiment-6	OPTICAL FIBRE: To determine the numerical aperture and acceptance angle of an optical fiber							
Experiment-7	LASER: To determine the wavelength of a given laser source by using diffraction grating method							
Experiment-8	NEWTON'S RINGS: To determine the radius of curvature of a given Plano convex lens by forming Newton's rings							
Experiment-9	THERMISTOR: To study the variation of resistance with respect to temperature using thermistor							
Experiment-10	Understanding the method of least squares - Torsional pendulum as an example							
Experiment-11	PLANCK'S CONSTANT: To determine value of planck's constant by measuring radiation in fixed spectral range							
Experiment-12	STEWART GEE'S EXPERIMENT: To study the variation of magnetic field along the axis of a circular coil and calculation of magnetic flux							
Note: Students have to perform any 8 experiments								
Reference Books:								
1. "Applied Physics Lab Manual"- Dr. Radhika Devi, Mr. A V Laxman Rao, N. Noel S. Balasubramanian, M.N Srinivasan "A Text book of practical Physics" – S Chand Publishers, 2017.								

ENGLISH LANGUAGE AND COMMUNICATION SKILLS

Course Code	Category	Hours /			Credits	Maximum Marks		
A6HS02	HSMC	L	T	P	C	CIE	SEE	Total
		0	0	3	1.5	40	60	100
English Language and Communication Skills Lab (ELCS) shall have two parts: a. Computer Assisted Language Learning (CALL) Lab b. Interactive Communication Skills (ICS) Lab								
Listening Skills Objectives 1. To enable students develop their listening skills to appreciate its role in the LSRW skills approach to language and improve their pronunciation. 2. To equip students with necessary training in listening so that they can comprehend the speech of people of different backgrounds and regions. Students should be given practice in listening to the sounds of the language, to be able to recognize them and find the distinction between different sounds, to be able to mark stress and recognize and use the right intonation in sentences. • Listening for general content • Listening to fill up information • Intensive listening • Listening for specific information								
Speaking Skills Objectives 1. To involve students in speaking activities in various contexts 2. To enable students express themselves fluently and appropriately in social and professional contexts • Oral practice • Describing objects/situations/people • Role play – Individual/Group activities • Just A Minute (JAM) Sessions								
Exercise-I								
CALL Lab: <i>Understand:</i> Listening Skill- Its importance – Purpose- Process- Types- Barriers- Effective Listening. <i>Practice:</i> Introduction to Phonetics – Speech Sounds – Vowels and Consonants – Minimal Pairs- Consonant Clusters- Past Tense Marker and Plural Marker- <i>Testing Exercises</i>								
ICS Lab: <i>Understand:</i> Spoken vs. Written language- Formal and Informal English. <i>Practice:</i> Ice-Breaking Activity and JAM Session- Situational Dialogues – Greetings – Taking Leave –Introducing Oneself and Others.								
Exercise-II								

CALL Lab:

Understand: Structure of Syllables – Word Stress– Weak Forms and Strong Forms – Stress pattern in sentences – Intonation.

Practice: Basic Rules of Word Accent - Stress Shift - Weak Forms and Strong Forms- Stress pattern in sentences – Intonation - *Testing Exercises*

ICS Lab:

Understand: Features of Good Conversation – Strategies for Effective Communication.

Practice: Situational Dialogues – Role Play- Expressions in Various Situations –Making Requests and Seeking Permissions - Telephone Etiquette.

Exercise-III**CALL Lab:**

Understand: Errors in Pronunciation-Neutralising Mother Tongue Interference (MTI).

Practice: Common Indian Variants in Pronunciation – Differences between British and American Pronunciation -Testing Exercises

ICS Lab:

Understand: Descriptions- Narrations- Giving Directions and Guidelines – Blog Writing

Practice: Giving Instructions – Seeking Clarifications – Asking for and Giving Directions – Thanking and Responding – Agreeing and Disagreeing – Seeking and Giving Advice – Making Suggestions.

Exercise-IV**CALL Lab:**

Understand: Listening for General Details.

Practice: Listening Comprehension Tests - Testing Exercises

ICS Lab:

Understand: Public Speaking – Exposure to Structured Talks - Non-verbal Communication- Presentation Skills.

Practice: Making a Short Speech – Extempore- Making a Presentation.

Exercise-V**CALL Lab:**

Understand: Listening for Specific Details.

Practice: Listening Comprehension Tests -Testing Exercises

ICS Lab:

Understand: Group Discussion

Practice: Group Discussion

Reference Books:

1. (2022). English Language Communication Skills – Lab Manual cum Workbook. Cengage Learning India Pvt. Ltd.
2. Shobha, KN & Rayen, J. Lourdes. (2019). Communicative English – A workbook. Cambridge University Press
3. Kumar, Sanjay & Lata, Pushp. (2019). Communication Skills: A Workbook. Oxford University Press
4. Board of Editors. (2016). ELCS Lab Manual: A Workbook for CALL and ICS Lab Activities. Orient Black Swan Pvt. Ltd.
5. Mishra, Veerendra et al. (2020). English Language Skills: A Practical Approach. Cambridge University Press

ENGINEERING WORKSHOP PRACTICES

I Semester (AERO, MECH & EEE)								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6ME03	ESC	L	T	P	C	CIE	SEE	Total
		0	1	2	2	40	60	100
WEEKS	TRADES FOR EXERCISES							
	Fitting							
Week 1	Filing Four Sides of MS Work piece							
Week 2	L Fit							
	Carpentry							
Week 3	Half Lap Joint							
Week 4	Dove Tail Joint							
	Tin Smithy							
Week 5	Prepare a Rectangular Tray							
Week 6	Prepare A Square Tin							
	Electrical							
Week 7	Series and Parallel Connection One Way Switch							
Week 8	Two Way Switch Connection Stair Case Wiring							
	Electronics							
Week 9	Soldering - Series Connection & Parallel Connection							
Week 10	Desoldering & Construction of Wheat stone bridge							
	Black Smithy							
Week 11	Blacksmithy : Round to S-Hook							
Week 12	Blacksmithy : Round to Chisel							
	TRADES FOR DEMONSTRATION & EXPOSURE							
Week 13	Introduction to Plumbing							
TEXT BOOKS								
1. Work shop Manual - P. Kannaiah/ K.L. Narayana/ Scitech. 2. Workshop Manual / K. Venugopal / Anuradha Publications.								
REFERENCE BOOKS:								
1. Hajra Choudhury S.K., Hajra Choudhury A.K.and Nirjhar Roy S.K., “Elements of Workshop Technology”, Media promoters and publishers private limited, Mumbai, Vol. I 2008 and Vol. II 2010. 2. Kalpakjian S. and Steven S. Schmid, “Manufacturing Engineering and Technology”, Pearson Education India Edition, 4 th edition, 2002.								

SEMINAR

I Semester Common to All								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6HS04	HSMC	L	T	P	C	CIE	SEE	Total
		0	0	2	1	50	-	50

OBJECTIVE:
Seminar is an important component of learning in an Engineering College, where the student gets acquainted with preparing a report & presentation on a topic.

PERIODICITY / FREQUENCY OF EVALUATION : Twice

PARAMETERS OF EVALUATION:

1. The seminar shall have topic allotted and approved by the faculty.
2. The seminar is evaluated for 50 marks for internal only.
3. The students shall be required to submit the rough drafts of the seminar outputs within one week of the commencement of the class work.
4. Faculty shall make suggestions for modification in the rough draft. The final draft shall be presented by the student within a week thereafter.
5. Presentation schedules will be prepared by Department in line with the academic calendar.

B. TECH II SEMESTER SYLLABUS

NUMERICAL METHODS AND INTEGRAL TRANSFORMS

II Semester: Common for All (ECE/EEE/MECH/AERO/ CSE/IT/AI&ML/CS/DS)

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6BS02	BSC	L	T	P	C	CIE	SEE	Total
		3	1	-	4	40	60	100
UNIT-I	INTERPOLATION AND CURVE FITTING							
INTERPOLATION: Finite differences: Forward, Backward and Central differences - Other difference operators and relations between them - Difference of a polynomial – Missing terms - Newton's forward interpolation, Newton's backward interpolation, Gauss's forward and backward interpolation formulae. Interpolation with unequal intervals – Lagrange's interpolation. CURVE FITTING: Method of least squares - Fitting a straight line, second degree parabola and non-linear curves of the form $y = a e^{bx}$, $y = a x^b$, $y = a b^x$ by the method of least squares.								
UNIT-II	NUMERICAL TECHNIQUES							
ROOT FINDING TECHNIQUES: Bisection method Regula falsi method, Iteration method and Newton Raphson method. NUMERICAL INTEGRATION : Trapezoidal rule - Simpson's one-third rule - Simpson's three-eighth rule. NUMERICAL SOLUTION OF ORDINARY DIFFERENTIAL EQUATIONS: Taylor's series method – Euler's - modified Euler's Method – Runge-Kutta method.								
UNIT-III	FOURIER SERIES							
Periodic function-Determination of Fourier Coefficients-Fourier Series-Even and Odd functions-Fourier series in arbitrary interval-Half range Fourier sine and cosine expansions.								
UNIT-IV	LAPLACE TRANSFORMS							
Laplace transforms of elementary functions- First shifting theorem - Change of scale property – Multiplication by t^n - Division by t – Laplace transforms of derivatives and integrals – Unit step function – Second shifting theorem – Periodic function – Evaluation of integrals by Laplace transforms – Inverse Laplace transforms- Method of partial fractions – Other methods of finding inverse transforms – Convolution theorem – Applications of Laplace transforms to ordinary differential equations.								
UNIT-V	FOURIER TRANSFORMS AND VECTOR DIFFERENTIATION							
Fourier Integral theorem (Statement only)-Fourier Sine and Cosine Integrals, Fourier Transforms, Cosine and Sine transforms, properties, Inverse transforms. Vector functions, vector differentiation, gradient, directional derivative, divergence, curl and scalar potential.								
Text Books:								
1. Ervin Kreyszig, Advanced Engineering Mathematics, 9 th Edition, John Wiley & Sons, 2006. 2. B.S.Grewal, Higher Engineering Mathematics, Khanna publishers, 36 th Edition, 2010.								
Reference Books:								
1. G.B.Thomas, calculus and analytical geometry,9 th Edition, Pearson Reprint 2006. 2. N.P Bali and Manish Goyal ,A Text of Engineering Mathematics,Laxmi publications,2008. 3. E.L.Ince, Ordinary differential Equations,Dover publications,1958.								

ENGINEERING CHEMISTRY

II Semester:: AERO, CSE, CSC, CSD, CSIT, CSM, EEE, ECE, IT, MECH								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6BS09	BSC	L	T	P	C	CIE	SEE	Total
		3	1	0	4	40	60	100
UNIT-I	WATER AND ITS TREATMENT							
Introduction to hardness of water – Estimation of hardness of water by complexometric method and related numerical problems. Potable water and its specifications - Steps involved in the treatment of potable water - Disinfection of potable water by chlorination and break - point chlorination. Defluoridation- Determination of F- ion by ion- selective electrode method. Boiler troubles: Sludges, Scales and Caustic embrittlement. Internal treatment of Boiler feed water -Calgon conditioning - Phosphate conditioning - Colloidal conditioning, External treatment methods -Softening of water by ion- exchange processes. Desalination of water – Reverse osmosis.								
UNIT-II	BATTERY CHEMISTRY AND CORROSION							
Introduction - Classification of batteries- primary, secondary and reserve batteries with examples. Basic requirements for commercial batteries. Construction, working and applications of: Zn-air and Lithium ion battery, Applications of Li-ion battery to electrical vehicles. Fuel Cells- Differences between battery and a fuel cell, Construction and applications of Methanol Oxygen fuel cell and Solid oxide fuel cell. Solar cells - Introduction and applications of Solar cells. Corrosion: Causes and effects of corrosion – theories of chemical and electrochemical corrosion –mechanism of electrochemical corrosion, Types of corrosion: Galvanic, water-line and pitting corrosion. Factors affecting rate of corrosion, Corrosion control methods- Cathodic protection – Sacrificial anode and impressed current methods.								
UNIT-III	POLYMERIC MATERIALS							
Definition – Classification of polymers with examples – Types of polymerization – addition (free radical addition) and condensation polymerization with examples – Nylon 6:6, Terylene Plastics: Definition and characteristics- thermoplastic and thermosetting plastics, Preparation, Properties and engineering applications of PVC and Bakelite, Teflon, Fiber reinforced plastics (FRP). Rubbers: Natural rubber and its vulcanization. Elastomers: Characteristics –preparation – properties and applications of Buna-S, Butyl and Thiokol rubber. Conducting polymers: Characteristics and Classification with examples-mechanism of conduction in trans-polyacetylene and applications of conducting polymers. Biodegradable polymers: Concept and advantages - Polylactic acid and poly vinyl alcohol and their applications.								
UNIT-IV	ENERGY SOURCES							
Introduction, Calorific value of fuel – HCV, LCV- Dulong's formula. Classification- solid fuels: coal –analysis of coal – proximate and ultimate analysis and their significance. Liquid fuels – petroleum and its refining, cracking types – moving bed catalytic cracking. Knocking – octane and cetane rating, synthetic petrol - Fischer-Tropsch's process; Gaseous fuels – composition and uses of natural gas, LPG and CNG, Biodiesel – Transesterification, advantages.								
UNIT-V	ENGINEERING MATERIALS							

Cement: Portland cement, its composition, setting and hardening.

Smart materials and their engineering applications

Shape memory materials- Poly L- Lactic acid. Thermo response materials- Polyacryl amides, Poly vinyl amides

Lubricants: Classification of lubricants with examples-characteristics of a good lubricants – mechanism of lubrication (thick film, thin film and extreme pressure)- properties of lubricants: viscosity, cloud point, pour point, flash point and fire point.

TEXT BOOKS:

1. Engineering Chemistry by P.C. Jain and M. Jain, Dhanpatrai Publishing Company, 2010
2. Engineering Chemistry by Rama Devi, Venkata Ramana Reddy and Rath, Cengage learning, 2016
3. A text book of Engineering Chemistry by M. Thirumala Chary, E. Laxminarayana and K. Shashikala, Pearson Publications, 2021.
4. 4. Textbook of Engineering Chemistry by Jaya Shree Anireddy, Wiley Publications

REFERENCE BOOKS:

1. Engineering Chemistry by Shikha Agarwal, Cambridge University Press, Delhi (2015)
2. 2. Engineering Chemistry by Shashi Chawla, Dhanpatrai and Company (P) Ltd. Delhi (2011)

ENGINEERING MECHANICS**I Year II Semester: (AERO & MECH)**

Course Code		Category		Hours / Week			Credits	Maximum Marks		
A6ME05		ESC		L	T	P	C	CIE	SEE	Total
				4	0	0	4	40	60	100
UNIT-I	INTRODUCTION TO ENGINEERING MECHANICS & EQUILIBRIUM OF PLANAR SYSTEM OF FORCES									
Introduction to engineering mechanics: Basic concepts, Classification of a force system, Parallelogram law, Resultant of coplanar concurrent force system, Moment of a force, its applications, Varignon's theorem, Resultant of coplanar non-concurrent force systems, Couples.										
Equilibrium of Planar System of forces: Equilibrium, Conditions of equilibrium, Free body diagrams, Lami's theorem.										
UNIT-II	FRICTION									
Friction: Introduction, Types of friction, Laws of friction, Coefficient of friction, Angle of friction, Angle of repose, Analysis of bodies on rough horizontal and inclined planes.										
Wedge friction, Ladder friction and Screw friction										
UNIT-III	CENTROID, CENTRE OF GRAVITY & MOMENT OF INERTIA									
Centroid & Centre of gravity: Introduction, Centroids & Centre of gravity of simple figures (from first principles), Centroid of composite sections, Centre of gravity of composite solids, Theorems of Pappus & Guldinus.										
Moment of Inertia: Area moment of inertia of plane sections (from first principles), Parallel axis theorem, Perpendicular axis theorem, Moment of inertia of standard sections and composite sections. Mass moment of inertia of rectangular plate, circular plate, cone, cylinder and sphere.										
UNIT-IV	KINEMATICS & KINETICS									
Kinematics of a particle: Rectilinear motion, Curvilinear motion, Rotary motion.										
Kinetics of rigid bodies: Analysis of connecting bodies, Kinetics of rigid body rotation, D' Alembert's principle, Work-Energy theorem, Impulse-Momentum principle, their applications.										
UNIT-V	VIRTUAL WORK & MECHANICAL VIBRATIONS									
Virtual work: Concept of virtual work, Principle of virtual work, its applications.										
Mechanical Vibrations: Basic terminology, Free and Forced vibrations, Resonance, Simple Harmonic Motion, Simple, Compound and Torsional Pendulums.										
Text Books:										
1. Reddy Vijay Kumar and K. Suresh Kumar, Singer's Engineering Mechanics 2. Engineering Mechanics/Timoshenko and D.H. Young, Mc Graw Hill Book Company 3. Bansal R.K, A Text book of Engineering Mechanics, Laxmi Publications										
Reference Books:										
1. Engineering Mechanics/Irving Shames/Prentice Hall 2. Engineering Mechanics/N.H Dubey/Tata Mc Graw Hill 3. A Text book of Engineering Mechanics/ R S Khurmi / S Chand										

PROGRAMMING FOR PROBLEM SOLVING

II Semester: Common for all Branches										
Course Code		Category		Hours / Week			Credits	Maximum Marks		
A6CS02		ESC		L	T	P	C	CIA	SEE	Total
				3	-	-	3	40	60	100
UNIT-I		INTRODUCTION - PROBLEM SOLVING AND ALGORITHMIC THINKING & INTRODUCTION TO C LANGUAGE								
Algorithm -Definition, Characteristics of Algorithm. Constituents of algorithms: - Sequence, Selection and Repetition. Algorithm with Example: Roots Of a Quadratic Equations, Minimum And Maximum Numbers of a Given Set, Given number is prime number or not, given integer is palindrome or not, etc. Flowchart/Pseudo Code with examples. Introduction To C Language: Structure of C Program, Data Types, data input and output statements, Operators, Precedence and Associativity of operators , Evaluation of Expressions, Type Conversions In Expressions.										
UNIT-II		CONTROL STRUCTURES AND ARRAYS								
Control structures: Decision statements; if and switch statement; Loop control statements: while, for and do while loops, Jump statements: break, continue, goto statements. Arrays: Concepts, One dimensional array, declaration and initialization of one dimensional arrays, two dimensional arrays, initialization and accessing, multi dimensional arrays										
UNIT-III		FUNCTIONS AND POINTERS								
Functions: Function definition, Types of Functions: User defined and built-in Functions, Advantages of User Defined Functions. Parameter passing in functions: Call by value, Call by reference, Passing arrays to functions, Recursion as a different way of solving problems. Example programs, such as Finding Factorial, Fibonacci series, Towers of Hanoi etc.. Storage classes. Pointers: Pointer basics, pointer arithmetic, pointers to pointers, generic pointers, array of pointers, Functions returning pointers, Dynamic memory allocation.										
UNIT-IV		STRINGS AND USER DEFINED DATA TYPES								
Strings: Arrays of characters, variable length character strings, inputting character strings, character library functions, String Handling Functions, Arrays Of Strings Structures and Unions: Structure definition, initialization, accessing structures, nested structures, arrays of structures, Structures and functions, Self-referential structures, unions, typedef, enumerations.										
UNIT-V		FILE HANDLING , SEARCHING AND SORTING								
File Handling: Command Line Arguments, File Modes, Basic File Operations Read, Write And Append, Example Programs. Random Access Using fseek, ftell and rewind Functions. Basic Searching And Sorting Algorithms: Linear and Binary Search, Bubble Sort ,Insertion Sort, Quick Sort.										
Text Books:										
1) B.A. Forouzan and R.F. Gilberg C Programming and Data Structures, Cengage Learning, (3rd Edition) 2) Byron Gottfried, "Programming with C", Schaum's Outlines Series, McGraw Hill Education, 3rd edition, 2017. 3) Programming in C E. Balagurusamy Edition 3 Publisher Tata McGraw-Hill Publishing, 1990										

Reference Books:

- 1) W. Kernighan Brian, Dennis M. Ritchie, "The C Programming Language", PHI Learning, 2nd Edition, 1988.
- 2) Yashavant Kanetkar, "Exploring C", BPB Publishers, 2nd Edition, 2003.
- 3) Schildt Herbert, "C: The Complete Reference", Tata McGraw Hill Education, 4th Edition, 2014.4. R. S. Bichkar, "Programming with C", Universities Press, 2nd Edition, 2012.
- 4) Dey Pradeep, Manas Ghosh, "Computer Fundamentals and Programming in C", Oxford University Press, 2nd Edition, 2006.
- 5) Stephen G. Kochan, "Programming in C", Addison-Wesley Professional, 4th Edition, 2014.

ELEMENTS OF AERONAUTICAL ENGINEERING

Semester								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AE01	ESC	L	T	P	C	CIA	SEE	Total
		-	-	3	2	40	60	100
UNIT-I	Basics of Aerodynamics							
Introduction-Anatomy of airships, gliders, airplane, helicopter, UAV - Forces on Flight Vehicles - Lift generation on the airplane - Drags and Types of Drag, Estimation of forces and moments on airfoil- Introduction to airfoils- nomenclature, Types of Airfoils- NACA Series and their applications, Laminar flow airfoils, Low Reynolds number airfoils								
UNIT-II	Basics of Aerospace structures							
Airship structural components- glider structural components, Airplane structural components-wing components- functions of each component-fuselage components-tail components and functions- landing gear components and functions, Helicopter structural components- rotor blades -tail rotors and functions								
UNIT-III	Aerospace Propulsion systems							
Introduction -principles of thrust generation- Thrust equation - propulsion used in airships reciprocating engine parts-propellers-gas turbine engine parts-functions-demonstration- engine starters- Introduction to rocket engines -								
UNIT-IV	Aircraft systems							
Introduction-hydraulic system components and functions- pneumatic systems- components and functions- electrical system-components-cabin-pressurization system-anti-iceing and deicing systems- landing gear system-cockpit display systems-Instruments-helicopter display system-instruments								
UNIT-V	Evolving technologies in Aeronautical Engineering							
Introduction-advanced control systems-components and functions, Applications of AI/MLI-Product Designing-Better Fuel Efficiency-Effective Supply Chain Management-Air Traffic Management-Identification of Threats- Applications of data science-Weather patterns and maintenance schedules. Data is collected, from flight takeoff to landing.Data patterns - analysis and algorithms for a safer flight Advanced flight deck display systems								
Text Books:								
1. Newman, D., Interactive Aerospace Engineering and Design, (with software and reference material on CD),McGraw-Hill, 2002, ISBN 0-07-112254-0. 2. Anderson, J.D., Introduction to Flight, fifth edition, Tata McGraw-Hill, 2007, ISBN: 0-07-006082-								
Reference Books:								
1. Barnard, R. H. and Philpot, D.R., Aircraft Flight, 3rd edition,Pearson, 2004,ISBN: 81-297-0783- 2. Machine Learning For Absolute Beginners by Oliver Theobald, 2020 3. A C Kermode, Flight Without Formulae, 5 th edition Pearson Education, 2008,ISBN:9788131713891								

I B. TECH I /II SEM: AERO, CSE, CSC, CSD, CSIT, CSM, EEE, ECE, IT, MECH

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A6BS10	BSC	L	T	P	C	CIE	SEE	Total
		1	0	2	1.5	40	60	100

LIST OF EXPERIMENTS

Experiment-1	Determination of total hardness of water by complexometric method using EDTA
Experiment-2	Determination of Alkalinity of given water sample
Experiment-3	Estimation of Chloride content of water by Argentometry.
Experiment-4	Estimation of the concentration of an acid by Conductometry
Experiment-5	Estimation of amount of ferrous ion by potentiometry using potassium dichromate
Experiment-6	Estimation of HCl by potentiometry
Experiment-7	Preparation of Bakelite
Experiment-8	Preparation of Thiokol Rubber
Experiment-9	Estimation of acid value of given lubricant oil
Experiment-10	Estimation of Viscosity of lubricant oil using Ostwald's Viscometer.
Experiment-11	Determination of rate of corrosion of mild steel in the presence and absence of inhibitor
Experiment-12	Estimation of Surface tension of a given liquids using Stalagmometer

VIRTUAL LABS

Experiment-13	Construction of Fuel cell and its working
Experiment-14	Smart materials for Biomedical applications
Experiment-15	Batteries for electrical vehicles
Experiment-16	Functioning of solar cell and its applications

Reference Books:

1. Lab manual for Engineering chemistry by B. Ramadevi and P. Aparna, S Chand Publications, New Delhi (2022)
2. Vogel's textbook of practical organic chemistry 5th edition
3. Inorganic Quantitative analysis by A.I. Vogel, ELBS Publications.
4. College Practical Chemistry by V.K. Ahluwalia, Narosa Publications Ltd. New Delhi (2007)

PROGRAMMING FOR PROBLEM SOLVING LAB

I - II Semester: Common to all branches

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6CS03	ESC	L	T	P	C	CIA	SEE	Total
		-	-	3	1.5	40	60	100

LIST OF EXPERIMENTS

Week-1 INTRODUCTION TO FLOWGORITHM

1. Installation and working of Flowgorithm Software.
2. Write and implement basic arithmetic operations using Flowgorithm – sum, average, product, difference, quotient and remainder of given numbers etc.
3. Draw a flowchart to calculate area of Shapes (Square, Rectangle, Circle and Triangle).
4. Draw a flowchart to find the sum of individual digits of a 3 digit number
5. Draw a flowchart to read input name, marks of 5 subjects of a student and display the name of the student, the total marks scored, percentage scored.
6. Draw a flowchart to find roots of a quadratic equation.
7. Draw a flowchart to find the largest and smallest among three entered numbers and also display whether the identified largest/smallest number is even or odd
8. Draw a flowchart to check whether the triangle is equilateral, isosceles or scalene triangle
9. Draw a flowchart to check whether a given number is palindrome or not.

Week-2 BASIC DATA TYPES

- 1) Write a C program to find division of 1 integer 1 float numbers.
- 2) Write a C program to find division of 2 integer numbers.
- 3) Write a C program to find average of (1int,1float) numbers.
- 4) Write a C Program to Swap Numbers without Using Temporary Variable
- 5) Write a C Program to Swap two Numbers Using Temporary Variable
- 6) Write a C Write a program to read the values of x, y and z and print the results of the following expressions in one line.
 - a) $(x+y+z) / (x-y-z)$
 - b) $(x+y+z) / 3$
 - c) $(x+y) * (x-y) * (y-z)$

Week-3 OPERATORS

- 1) Write a C program to convert temperature from Fahrenheit to Celsius and vice versa ($c=(fh-32)/1.8$)
- 2) Write a C program to find area and perimeter of a circle. ($area=\pi r^2$ perimeter= $2\pi r$)
- 3) Write a C program to calculate area and perimeter of a right angled triangle.
 - a. ($Area=1/2*b*h$ perimeter= $w+h+sqrt(w^2+h^2)$)
- 4) Find the sum of natural numbers 1 to n.(read n as input) (Use formula $sum=n(n+1)/2$)
- 5) Write a C program to calculate Simple interest ($SI=PTR/100$)
- 6) Write a C program to calculate area and perimeter of a rectangle. Area= $l*b$ Perimeter= $2*(l+b)$

- 7) Write a C program to calculate the value of the third angle of a triangle if two angles are given as input. ($a+b+c=180$)
- 8) Write a C program to read the consumer number and number of units consumed and the cost per unit and print the amount to be paid. ($\text{Amt} = \text{num of units} \times \text{cost}$)
- 9) Write a C Program to calculate area and perimeter of a triangle.
 - a. Perimeter = $(a+b+c)$
 - b. $s = (a+b+c)/2$
 - c. Area = $\text{sqrt}(s(s-a)(s-b)(s-c))$
- 10) Write a C program to read five Subject marks and find the average.
- 11) Write a C program to Calculate Compound interest ($CI = p(1+r/100)^n$)

Week-4**CONDITIONAL STATEMENTS**

1. Write a C program to find largest and smallest of given numbers.
2. Write a C program which takes two integer operands and one operator from the user (+, -, *, /, % use switch)
3. Write a program to compute grade of students using if else ladder. The grades are assigned as followed:

$\text{marks} < 50$ F
 $50 \leq \text{marks} < 60$ C
 $60 \leq \text{marks} < 70$ B
 $70 \leq \text{marks}$ B+
 $80 \leq \text{marks} < 90$ A
 $90 \leq \text{marks} \leq 100$ A+
4. Write a C program to whether given year leap year or not.
5. Write a C program to find whether given triangle is scalene or isosceles or equilateral.

Week-5**LOOPING STATEMENTS**

- 1) Write a C program to find Sum of individual digits of given integer
- 2) Write a C program to generate first n terms of Fibonacci series
- 3) Write a C Program to find the Sum of Series $\text{SUM} = 1 - x^2/2! + x^4/4! - x^6/6! + x^8/8! - x^{10}/10!$
- 4) Write a C program to print the Fibonacci sequence up to given value of n.
- 5) Write a C program to print the multiplication table to the given value of n.
- 6) Write a C program to check whether given number is palindrome or not
- 7) Write a C program to check whether given number is perfect or not

Week-6**NESTED LOOPING STATEMENTS**

- 1) Write a C program to generate prime numbers between 1 and n
- 2) Write a C program to generate Pascal's triangle.
- 3) Write a C program to generate the following pyramid of numbers.

```

          1
        1 3 1
      1 3 5 3 1

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- 4) Write a C program to generate the pattern

<pre> ***** ***** ***** ***** ***** </pre>	
Week-7	ARRAYS
- Write a C Program to implement following searching methods - Binary Search - Linear Search - Write a C program to find largest and smallest number in a list of integers - Write a C program - To add two matrices - To multiply two matrices - Write a C program to find Transpose of a given matrix	
Week-8	FUNCTIONS
- Write a C program to find the factorial of a given integer using non recursive functions - Write a C program to find GCD of given integers using non recursive functions - Write a C Program to find the power of a given number using non recursive functions - Write a C program to find sum of natural numbers using non recursive function. - Write a C program to reverse a given integer number using non recursive functions - Write a C Program to find binary equivalent of a given decimal number using recursive functions. - Write a C Program to print Fibonacci sequence using recursive functions. - Write a C Program to find LCM of 3 given numbers using recursive functions - Write a C program to find the factorial of a given integer using recursive functions - Write a C program to print fibonacci series till n terms using recursion.	
Week-9	STRINGS
- Write a C program using to Insert a sub string into a given main string from a given position - Write a C program using to Delete n characters from a given position in a string - Write a C program to determine if given string is palindrome or not - Write C Programs to demonstrate the following string handling functions. - strcat() - strcmp() - strrev() - strcpy() - strlen() - strstr() - strncpy() - strncat() - strncmp()	
Week-10	POINTERS

- 1) Write a C program to read the elements of 1-d array using pointers and print them in reverse order using pointers.
- 2) Write a C Program to read two elements dynamically using malloc() function and interchange the two numbers using call by reference.
- 3) Write a C Program to read and print the elements of 1-D array using calloc() memory allocation function and reallocate memory for the array by increasing the size of the array, read and print the elements of reallocated array.
- 4) Write a C Program to print 2-D array using pointers.

Week-11	STRUCTURES
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- 1) Write a C Program using functions to
 - a) Reading a complex number
 - b) Writing a complex number
 - c) Add two complex numbers
 - d) Multiply two complex numbers

Note: represent complex number using structure
- 2) Write a C program to read employee details employee number, employee name, basic salary, hra and da of n employees using structures and print employee number, employee name and gross salary of n employees.

Week-12	FILES
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- 1) Write a C program to read and print the content of a file.
- 2) Write a C program copy the content of one file to another file
- 3) Write a C program to merge two file into third file.
- 4) Write a C Program to find the number of lines in a text file

Text Books:

- 1) Riley DD, Hunt K.A. Computational Thinking for the Modern Problem Solver. CRC press, 2014 Mar 27.
- 2) B.A. Forouzan and R.F. Gilberg C Programming and Data Structures, Cengage Learning, (3rd Edition) Yashavant Kanetkar, "Let Us C", BPB Publications, New Delhi, 13th Edition, 2012.

Reference Books:

- 1) Ferragina P, Luccio F. Computational Thinking: First Algorithms, Then Code. Springer; 2018
- 2) King KN, "C Programming: A Modern Approach", Atlantic Publishers, 2nd Edition, 2015.
- 3) Kochan Stephen G, "Programming in C: A Complete Introduction to the C Programming Language", Sam's Publishers, 3rd Edition, 2004.
- 4) Linden Peter V, "Expert C Programming: Deep C Secrets", Pearson India, 1st Edition, 1994.

ENVIRONMENTAL SCIENCE

II SEMESTER: COMMON TO ALL BRANCHES

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A6BS11	MC	L	T	P	C	CIE	SEE	Total
		2	0	0	0	50	--	50
UNIT-I	ECOSYSTEMS							
Ecosystems: Definition, Scope and Importance of ecosystem. Classification, structure and function of an ecosystem, Food chains, food web and ecological pyramids.								
Flow of energy, Biogeochemical cycles, Bioaccumulation, Bio magnification, ecosystem value, services and carrying capacity.								
UNIT-II	NATURAL RESOURCES							
Natural Resources: Classification of Resources: Living and Non-Living resources, water resources: use and over utilization of surface and ground water, floods and droughts, Dams: benefits and problems-case studies. Mineral resources: use and exploitation, environmental effects of extracting and using mineral resources. Land resources: Forest resources.								
Energy resources: Growing energy needs, renewable and non-renewable energy sources, use of alternate energy source, case studies.								
UNIT-III	BIODIVERSITY AND BIOTIC RESOURCES							
Biodiversity and Biotic Resources: Introduction, Definition, genetic, species and ecosystem diversity. Value of biodiversity; consumptive use, productive use, social, ethical, aesthetic and optional values.								
India as a mega diversity nation: Hot spots of biodiversity. Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts; conservation of biodiversity: In-Situ and Ex-situ conservation. National Biodiversity Act-Case studies.								
UNIT-IV	ENVIRONMENTAL POLLUTION AND CONTROL TECHNOLOGIES							
Environmental Pollution and Control Technologies: Environmental Pollution: Classification of pollution, Air Pollution, Water Pollution drinking water quality standards. Soil Pollution Impacts of modern agriculture, Noise Pollution Health hazards, standards. Concepts of bioremediation.								
Global Environmental Problems and Global Efforts: Ozone depletion and Ozone depleting substances (ODS) Concepts of Bioremediation International conventions / Protocols: Earth summit, Kyoto protocol and Montréal Protocol; NAPCC-GoI Initiatives, COP 24, COP25.								
UNIT-V	ENVIRONMENTAL POLICY, LEGISLATION & EIA							
Environmental Policy, Legislation & Environmental Impact Assessment (EIA): Environmental Protection act, Legal aspects Air Act1981, Water Act, Forest Act, Wild life Act, Municipal solid waste management, biomedical waste management hazardous waste management and handling rules.								
Environmental Impact Assessment: EIA structure, methods of baseline data acquisition. Strategies for risk assessment, Towards Sustainable Future: Concept of Sustainable Development, Urban Sprawl, Concept of Green Building.								
TEXT BOOKS:								
1. Textbook of Environmental Studies for Undergraduate Courses by Erach Bharucha for University Grants Commission.								
2. Environmental Studies by R. Rajagopalan, Oxford University Press.								

B. TECH III SEMESTER SYLLABUS

AIRPLANE PERFORMANCE

III Semester								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AE02	PCC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
UNIT-I	INTRODUCTION TO AIRPLANE PERFORMANCE							
Role and design mission of an aircraft specification of the performance requirements and mission profile. Importance of performance analysis, estimation and measurements .Scheduled performance and operational performance of aircraft. International standard Atmosphere. Off - design atmosphere. Measurements of air data. Air data computers.								
UNIT-II	AIRPLANE CRUISE PERFORMANCE							
Thrust required and thrust available curves-Minimum drag speed, minimum power speed. Aerodynamic relationships for a parabolic drag polar. maximum & minimum speeds in level flight -Range and Endurance of aircraft with thrust producing engines and with power producing engines. Cruise techniques: constant angle of attack, constant Mach number, constant Mach number methods, comparison of performance. Effect of weight, altitude and temperature on cruise performance.								
UNIT-III	AIRPLANE CLIMB ,DESCENT AND MANEUVER PERFORMANCE							
Importance of climb and descent performance- safety considerations. Climb and descent techniques, generalized performance analysis for thrust producing, power producing and mixed power plants, maximum climb gradient, climb rate. Descent performance in aircraft operations. Effect of wind on climb and descent performance.manoeuvre performance- v-n diagram -level turn-pull up maneuver -pull down manoeuvre								
UNIT-IV	AIRPLANE TAKE-OFF AND LANDING PERFORMANCE AND FUEL PLANNING							
Estimation of take off distances. effect on the take – off distance, of weight wind runaway conditions, ground effect. Take off performance safety factors. Estimation of landing distances, the discontinued landing baulked landing air safety procedures and requirements on performance. Fuel Planning : Fuel planning fuel requirements trip fuel, fuel reserves, tankering. Purpose of performance measurements in flight.Flight testing Principle performance variables weight, altitude and ambient temperature (WAT).								
UNIT-V	APPLICATION OF PERFORMANCE DATA							
Performance summary and fleet selection-the block performance,payload – range diagram. Route analysis and optimization. Operational analysis procedure. Operational performance data for flight planning, take off field performance runway correction chart, aircraft datum performance (WAT) chart, determination of the maximum takeoff weight								
Text Books:								
1. Eshelby M. E (2000), <i>Aircraft performance: Theory and Practice</i> , AIAA Education Series, USA. 2. Anderson J. D. (2011), <i>Aircraft Performance and Design</i> , international edition, McGraw Hill, New Delhi.								
Reference Books:								
1. Mc Cormic B. W.(2010), <i>Aerodynamics, Aeronautics and Flight Mechanics</i> , 2 nd edition, Wiley India Ltd. India 2 Yethout (2003), <i>Introduction to Aircraft Flight Mechanics</i> , AIAA Education Series, USA								

SOLID MECHANICS FOR AERONAUTICS

III Semester								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AE03	PCC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
UNIT-I	SIMPLE STRESSES AND STRAINS							
SIMPLE STRESSES AND STRAINS: Elasticity and plasticity, Types of stresses and strains, Hooke's law, stress-strain diagram for mild steel, Working stress, Factor of safety, Lateral strain, Poisson's ratio and volumetric strain, Elastic constants and the relationship between them, stepped bars, composite bars, Temperature stresses. Strain energy-suddenly applied and impact loads								
UNIT-II	SHEAR FORCE AND BENDING MOMENT							
SHEAR FORCE AND BENDING MOMENT: Introduction to beam, Shear force and bending moment, Relation between Shear Force and Bending Moment. Shear Force and Bending Moment diagrams for cantilever, simply supported and overhanging beams subjected to point loads and U.D.L., Point of contra flexure.								
UNIT-III	REDUNDANT STRUCTURES							
DEFLECTION OF BEAMS: Double integration methods, Determination of slope and deflection for cantilever and simply supported beams subjected to point loads. Indeterminate structure, order of redundancy, moment area method, analysis of various types of beams, three moment equation (Clapeyron theorem) of a continuous beam.Principle of virtual Displacements and principle of Virtual Force Castigliano's theorems, maxwell's reciprocal theorem and unit load method								
UNIT-IV	FLEXURAL STRESSES and TORSIONAL STRESSES							
FLEXURAL STRESSES: Theory of simple bending, Assumptions, Derivation of bending equation: $M/I = f/y = E/R$, Neutral axis, Determination bending stresses-I, T sections- section modulus of rectangular and circular sections (Solid and Hollow), TORSION OF CIRCULAR SHAFTS: Theory of pure torsion - derivation of Torsion equations: $T/J = q/r = G\theta/L$ - Assumptions made in the theory of pure torsion - Torsional moment of resistance - Polar section modulus - Power transmitted by shafts								
UNIT-V	PRINCIPAL STRESSES AND STRAINS & FAILURE THEORIES							
PRINCIPAL STRESSES AND STRAINS: Introduction - Stresses on an inclined section of a bar under axial loading - compound stresses - Normal and tangential stresses on an inclined plane for biaxial stresses - Two perpendicular normal stresses accompanied by a state of simple shear - Mohr's circle of stresses Various theories of failure - Maximum Principal Stress Theory, Maximum Principal Strain Theory, Strain Energy and Shear Strain Energy Theory (Von Mises Theory).								
Text Books:								
1.Ramamrutham. S (2012), Strength of materials, 17 th edition, Dhanpat Rai Publications, Engineering Mechanics/Timoshenko and D.H. Young, Mc Graw Hill Book Company New Delhi, India. 2.Dr.Bansal R.K(2007), Strength of materials, 10 th edition,Laxmi Publications,Hyderabad								
Reference Books:								
1. Ryder G. H (2007), Strength of materials, 3 rd edition, Macmillan, New Delhi, India. 2. Bhavikathi S. S (2008). Strength of materials. 3 rd edition. Vikas Publishing House. New Delhi. India								

AERODYNAMICS

III Semester								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AE04	PCC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
UNIT-I	FUNDAMENTALS OF FLUID MECHANICS							
Flow properties, Hydro static law, Piezometers, Simple and differential manometers, Velocity Potential and Stream function -Stream line, path line, streak line, stream tube, classification of flows, steady, unsteady, uniform, non-uniform, laminar, turbulent, rotational, irrotational flows, Continuity equation, Momentum equation -Euler's and Bernoulli's equation, Navier Stokes equation (explanation only),								
UNIT-II	ELEMENTARY FLOWS							
Introduction to Elementary Flows-Uniform flow, source flow, sink flow, doublet flow and vortex flow, non-lifting and lifting flow over a circular cylinder. Kutta- Joukowski theorem and Magnus Effect, D'Alembert's paradox								
UNIT-III	AIRFOIL AND FINITE WING THEORY							
THIN AIRFOIL THEORY: Vortex Filament, The vortex sheet, Kutta condition and Kelvin 's circulation theorem. Classical thin airfoil theory: symmetric and cambered airfoil. aerodynamic centre, centre of pressure, estimation of coefficient of lift and pitching moment.								
Finte Wing Theory: Downwash, induced drag, Biot-Savart's law and Helmholtz's theorem. Prandtl's classical lifting line theory and fundamental equations. Elliptic and general lift distribution over finite unswept wings, effect of aspect ratio, taper and thickness to chord ratio, Subsonic flow past swept and delta wings.								
Assignment/Project submission: Program to calculate lift over wing Prandtl's classical lifting line theory								
UNIT-IV	APPLIED AERODYNAMICS-I							
Flow regimes -Flow over Wing and airfoil, Lift augmentation and Drag Reduction methods - Flaps, slats, slots, winglets, Leading edge root extensions, Large Eddy Breakup device, Co-flow jet, Cuffs and vortex generators Circulation control, strakes. Critical Mach number, drag divergence Mach number, supercritical airfoils, area rule. Drag augmentation methods – spoilers, Air brakes.								
UNIT-V	APPLIED AERODYNAMICS-II							
Propellers: Airscrew geometry, Froude Momentum Theory, Thrust Co-eff, Torque Co-eff, Power Co-eff, Efficiency, Activity factor, Blade element theory.								
Boundary layer Theory:								
Boundary layers, Laminar flow and turbulent flow, Boundary layer thickness, displacement and momentum thickness, boundary layer separation.								
Assignment/Project submission: Prepare models for wings different flap positions, and leading-edge devices								
Text Books:								
1. John. M. Cimbala Yunus A. Cengel (2019), Fluid Mechanics: Fundamentals and Applications, 4 th edition, McGraw-Hill, ISBN: 9353166217								
2. Anderson J.D. (2011), Fundamental of Aerodynamics, 5 th edition, McGraw-Hill International Edition, New York.								
Reference Books:								
1. L. J. Clancy, Aerodynamics, 1/e, Shroff Publications, 2006								
2. J. J. Bertin and R. Cummings, Aerodynamics for Engineers, 6/e, Pearson, 2013.								

AERO THERMODYNAMICS

III Semester								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AE05	ESC	L	T	P	C	CIA	SEE	Total
		3	1	-	4	40	60	100
UNIT-I	BASIC CONCEPT AND FIRST LAW OF THERMODYNAMICS							
Basic Concepts: Macroscopic and Microscopic viewpoints, Quasi static Process, various flow and non-flow process, energy in State and in Transition, Work and Heat, Point and Path function. Zeroth Law of Thermodynamics, First law of Thermodynamics, First law applied to a Process, applied to a flow system.								
UNIT-II	SECOND LAW OF THERMODYNAMICS AND ENTROPY PRINCIPLES							
LIMITATIONS OF THE FIRST LAW: Thermal Reservoir, Heat Engine, Heat pump Refrigerator. Parameters of performance, Second Law of Thermodynamics-Kelvin Planck and Clausius Statements and their Equivalence / Corollaries, PMM of Second kind, Carnot's principle, Carnot cycle and its specialties, Clausius Inequality, Entropy, Principle of Entropy Increase, Elementary Treatment of the Third Law of Thermodynamics.								
UNIT-III	VAPOUR AND GAS POWER CYCLES							
VAPOUR POWER CYCLES: P-V-T surfaces, T-S and h-s diagrams, Mollier Charts, Phase transformations, Dryness Fraction. Various thermodynamic processes and energy transfer. Simple Rankine cycle. GAS POWER CYCLES: Otto, Diesel and Brayton cycles, Description and representation on P-V and T-S diagram, Thermal Efficiency, Mean Effective Pressures on Air standard basis, comparison of Cycles. Basics of Jet propulsion.								
UNIT-IV	GAS TURBINE ENGINE COMPRESSORS							
Axial Flow Compressors: Elementary theory of axial flow compressor, Velocity triangles, Degree of reaction. Flow through an Axial flow Compressor stage; Losses in axial flow compressor stage; Compressor blade design, Stage losses, Characteristics of Blade - single and multi-stage Axial compressor.								
UNIT-V	GAS TURBINE ENGINE TURBINES							
Turbines: Introduction; Turbine stage; Turbine Blade 2-D (cascade) analysis Work Done; Degree of Reaction; Losses and Efficiency; Flow Passage; Subsonic, transonic and supersonic turbines, Multi-staging of Turbine, Turbine Cooling.								
Text Books:								
1. Engineering Thermodynamics, P.K. Nag, 6 th Edition, Mc Graw Hill Education. 2. Gas Turbines, V Ganesan, 3 rd Edition, Mc Graw Hill Education (India) Private Limited								
Reference Books:								
1. Fundamentals of engineering thermodynamics, Rathakrishnan. E, 2 nd Edition, Prentice hall of India Pvt Ltd., 2006. 2. Gas Turbine Theory, H.I.H Saravanamuttoo, G.F.C Rogers and H. Colen, 7 th Edition, Pearson								

BASIC ELECTRICAL AND ELECTRONICS ENGINEERING

III Semester								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE60	ESC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
UNIT-I	ELECTRICAL CIRCUITS							
Basic definitions-Ohm’s Law, types of elements, types of sources , Kirchhoff’s Laws – simple problems., series & parallel resistive networks with DC excitation, star to delta and delta to star transformations								
UNIT-II	DC MACHINES							
Principle of Operation of DC Motor, types of DC motor, Torque equation & Losses and problems. DC Generator construction and working Principle, EMF Equation types of generators and problems.								
UNIT-III	AC MACHINES							
Working principle and Construction of transformer, Emf Equation & problems. Principle operation of 3-phase induction motor, slip and torque Equation, Torque –slip characteristics & problems.								
UNIT-IV	DIODE AND ITS CHARACTERISTICS							
PN JUNCTION DIODE: Operation of PN junction Diode: forward bias and reverse bias, Characteristics of PN Junction Diode – Zener Effect – Zener Diode and its Characteristics. Rectifiers, Half wave, Full wave and bridge Rectifiers –capacitor filters, inductor filters								
UNIT-V	TRANSISTORS							
Bipolar Junction Transistor - NPN & PNP Transistor, CB, CE, CC Configurations and Characteristics.								
Text Books:								
1. Basic Electrical Engineering by M.S.Naidu and S.Kamakshaiah TMH 2. Electronic Devices and circuits by J.Millman, C.C.Halkias and Satyabrata Jit 2ed.,								
Reference Books:								
1. Muthusubramanian R, Salivahanan S and Muraleedharan K A, “Basic Electrical, Electronics and Computer Engineering”,Tata McGraw Hill, Second Edition, (2006). 2. Nagsarkar T K and Sukhija M S, “Basics of Electrical Engineering”, Oxford press (2005). 3. Mehta V K, “Principles of Electronics”, S.Chand & Company Ltd, (1994). 4. Mahmood Nahvi and Joseph A. Edminister, “Electric Circuits”, Schaum’ Outline Series, McGraw Hill, (2002).								

SOLID MECHANICS FOR AERONAUTICS LAB

III Semester								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AE06	ESC	L	T	P	C	CIA	SEE	Total
		-	-	3	1.5	40	60	100
LIST OF EXPERIMENTS								
MOS LAB								
1. Direct Tension Test on UTM 2. Deflection test on Simple supported Beam 3. Deflection test on Cantilever Beam 4. Torsion Test 5. Brinell hardness test 6. Compression test on cube 7. Test on springs 8. Impact Strength-Izod Test 9. Impact Strength-Charpy Test 10. Shear Test on UTM								
Reference Books:								
1. R. K. Bansal (2011), A Textbook of Fluid Mechanics and Hydraulic Machines, 10th edition,Laxmi Publications, New Delhi, India. 2. Ramamrutham. S (2012), Strength of materials, 17th edition, Dhanpat Rai Publications,								

BASIC ELECTRICAL AND ELECTRONICS ENGINEERING LAB

III Semester								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE61	ESC	L	T	P	C	CIA	SEE	Total
		0	0	2	1	40	60	100
LIST OF EXPERIMENTS								
PART A : ELECTRICAL								
1	a. Identification, Specifications, Testing of R, L, C Components (Color Codes), Bread Boards, Diodes, BJTs. b. Verification of Ohms law							
2	Verification of KCL and KVL.							
3	Magnetization characteristics of DC shunt generator (Determination of critical field resistance and critical speed)							
4	Swinburne's test on dc shunt motor (Predetermination of efficiencies)							
5	OC and SC tests on Single-phase transformer.							
6	Brake test on 3-phase Induction motor.							
	PART B : ELECTRONICS							
7	Study and operation of Function Generator Regulated Power Supply (RPS) CRO.							
8	PN Junction Diode Characteristics (Forward bias, Reverse bias)							
9	Zener Diode Characteristics							
10	Rectifier without Filters (Full wave & half wave).							
11	Rectifier with Filters (Full wave & half wave).							
12	Transistor CE Characteristics (Input and Output)							
Text Books:								
1. Basic Electrical Engineering by <i>M.S.Naidu and S.Kamakshaiah</i> TMH 2. Electronic Devices and circuits by <i>J.Millman, C.C.Halkias and Satyabrata Jit</i> 2ed.,								
Reference Books:								
1. Muthusubramanian R, Salivahanan S and Muraleedharan K A, "Basic Electrical, Electronics and Computer Engineering", Tata McGraw Hill, Second Edition, (2006). 2. Nagsarkar T K and Sukhija M S, "Basics of Electrical Engineering", Oxford press (2005). 3. Mehta V K, "Principles of Electronics", S.Chand & Company Ltd, (1994). 4. Mahmood Nahvi and Joseph A. Edminister, "Electric Circuits", Schaum' Outline Series, McGraw Hill, (2002).								

AERODYNAMICS LAB

III Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AE07	PCC	L	T	P	C	CIA	SEE	Total
		-	-	3	1.5	40	60	100

LIST OF EXPERIMENTS

1. Calibration of Venturi meter
 2. Determination of Coefficient of discharge for a small orifice by a constant head method.
 3. Determination of Coefficient of loss of head in a sudden contraction and friction factor.
 4. Verification of Bernoulli's equation
 5. Calibration of subsonic wind tunnel.
 6. Pressure distributions over smooth and rough cylinder.
 7. Pressure distribution over cambered airfoils
 8. Flow visualization studies over sphere.
 9. Flow visualization studies over cylinders.
 10. Flow visualization studies in low-speed flows over airfoil with different angle of incidence.
 11. Drags estimation on airfoils
 12. Drags estimation on blunt bodies
- Note: Ten experiments should be performed.

Reference Books:

1. John. M. Cimbala Yunus A. Cengel (2019), Fluid Mechanics: Fundamentals and Applications, 4th edition, McGraw-Hill, ISBN: 9353166217
2. Jewel B. Barlow(1999), Low-Speed Wind Tunnel Testing, 3rd edition, Wiley-Interscience, ISBN: 0471557749

INTELLECTUAL PROPERTY RIGHTS

III Semester								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6HS13	HSMC	L	T	P	C	CIA	SEE	Total
		2	-	-	-	40	60	100
UNIT-I								
Introduction to Intellectual property: Introduction, types of intellectual property, international organizations, agencies and treaties, importance of intellectual property rights.								
UNIT-II								
Trade Marks: Purpose and function of trademarks, acquisition of trade mark rights, protectable matter, selecting and evaluating trade mark, trade mark registration processes.								
UNIT-III								
Law of copy rights: Fundamental of copy right law, originality of material, rights of reproduction, rights to perform the work publicly, copy right ownership issues, copy right registration, notice of copy right, international copy right law. Law of patents: Foundation of patent law, patent searching process, ownership rights and transfer.								
UNIT-IV								
Trade Secrets: Trade secrete law, determination of trade secrete status, liability for misappropriations of trade secrets, protection for submission, trade secrete litigation. Unfair competiitiion: Misappropriation right of publicity, False advertising.								
UNIT-V								
New development of intellectual property: new developments in trade mark law; copy right law, patent law, intellectual property audits. International overview on intellectual property, international - trade mark law, copy right law, international patent law, international development in trade secrets law.								
Text Books:								
1. Intellectual property right, Deborah, E. Bouchoux, cengage learning. 2. Intellectual property right - Unleashing the knowledge economy, prabuddha ganguli, Tata Mc Graw Hill Publishing Company Ltd.								
Reference Books:								
1. Deborah. E. Bouchoux (2009), Intellectual property, Cengage learning, India. 2. Deborah. E. Bouchoux (2001), Protecting your companies intellectual property, AMACOM, USA. 3. Prabuddaganguli (2003), Intellectual property right, Tata McGraw Hill Publishing company Ltd., India. 4. Robert Hisrich, Michael P. PEter, Dean A. Shepherd (201), Entrepreneurship, Tata McGraw Hill., India.								

B.TECH IV SEMESTER SYLLABUS

IOT BASED DRONE SYSTEMS

IV Semester								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AE08	ESC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100
UNIT-I	INTRODUCTION TO IOT							
What is the IoT and why is it important? Elements of an IoT ecosystem, Technology drivers, Business drivers, Trends and implications, Overview of Governance, Privacy and Security Issues.								
UNIT-II	IOT ARCHITECTURE							
IOT ARCHITECTURE - IoT Open source architecture (OIC)- OIC Architecture &Design principles- IoT Devices and deployment models- IoTivity : An Open source IoT stack - Overview- IoTivity stack architecture- Resource model and Abstraction								
UNIT-III	WEB OF THINGS							
WEB OF THINGS - Web of Things versus Internet of Things – Two Pillars of the Web – Architecture Standardization for WoT– Platform Middleware for WoT – Unified Multitier WoT Architecture – WoT Portals and Business Intelligence								
UNIT-IV	UAV SUBSYSTEMS							
Mission Planning and Control Station- Types, Physical Configuration, Planning and Navigation, MPCS Interfaces, Modes of control, piloting and controlling mission, Autopilot system Launch Systems- Basic Considerations, launch Methods for Fixed, rotary wing UAV Recovery Systems								
UNIT-V	Interoperability in IoT and Introduction to Arduino Programming							
Sensing, Actuation, Basics of Networking,Communication Protocols ,Sensor Networks, Machine-to-Machine, Integration of Sensors and Actuators with Arduino								
Text Books:								
1. Honbo Zhou, “The Internet of Things in the Cloud: A Middleware Perspective”, CRC Press,2012. 2. Dieter Uckelmann, Mark Harrison, Michahelles, Florian (Eds), “Architecting the Internet of Things”, Springer, 20112.								
Reference Books:								
1. CunoPfister, Getting Started with the Internet of Things, O'Reilly Media, 2011, ISBN: 978-1-4493-9357-1 2. Skafidas, "Microcontroller Systems for a UAV", KTH, TRITA-FYS 2002:51 ISSN 0280-316X.34, 2002. 3.Vijay Madiseti and ArshdeepBahga, “Internet of Things (A Hands-on-Approach)”, 1st Edition,VPT, 2014 4.Olivier Hersent, David Boswarthick, Omar Elloumi , “The Internet of Things – Key applicationsand Protocols”, Wiley, 2012								

AEROSPACE PROPULSION

IV Semester								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AE09	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
UNIT-I	BACKGROUND AND REQUIREMENTS							
Introduction, Motion of bodies in space, parameters describing motion of bodies, frame of reference Impulse, force, universal law for gravitational force, motion in rotating frame of reference, pseudo- centrifugal force Orbits, orbit velocities, orbital period, geosynchronous and geostationary orbits, eccentricity and inclination, polar, sun-synchronous and other orbits Energy requirements for orbit, escape velocity								
UNIT-II	NOZZLES AND ROCKETS							
NOZZLES: Rocket Nozzles:Expansion of gases from high pressure chamber, efflux velocity, classification, Performance losses in nozzles, Under-and Over- expansion nozzles (for very large pressure ratios),Performance loss in nozzles, mass flow rates and characteristic velocity role for interplanetary and deep space missions ROCKETS: Introduction -Rocket principle and Tsiolkovsky Rocket equation Mass ratio of rocket, desirable parameters to achieve high velocities, propulsive efficiency Performance parameters of a rocket, staging and clustering								
UNIT-III	SOLID PROPELLANT ROCKETS							
Solid propellant rockets: Introduction, classification Propellant grain configurations and its loading strategies: design of solid propellant rocket, Ignition of solid propellant rockets, ignition problems and solutions Characteristic burn times and action time. Liners, Insulators and inhibitors- Thrust developed by rocket , Net-and Gross- thrust, specific impulse- Calculation of heat of combustion, temperature, molecular mass and rocket performance parameters								
UNIT-IV	LIQUID PROPELLANT ROCKETS AND CRYOGENICS							
LIQUIDPROPELLANT ROCKETS: Introduction to liquid propellant rockets, propellant feed systems, Gas Pressure feed system, Propellant properties, Liquid oxidizers and fuels Cryogenics : Fundamentals of cryogenic s – components-Cryogenic fuels/propellant-Cryogenic safety aspects-Basic cryomodule design considerations- Next generation cryogenic engine								
UNIT-V	COMBUSTION INSTABILITY IN ROCKETS AND ELECTRICAL ROCKETS							
COMBUSTION INSTABILITY IN ROCKETS: Combustion instability in rockets; illustration through examples, bulk and wave modes of instability, Modeling of bulk mode of instability in solid and liquid propellant rockets ELECTRICAL ROCKETS Introduction- electrical and magnetic fields, electro-thermal, arc-jet, electrostatic and electromagnetic thrusters								
Text Books:								
1. Ramamurthi, K., Rocket Propulsion, Macmillan (in press) 2009 2.Sutton, G. P. and Biblarj, O. Rocket propulsion elements, 7th Ed., New York: Wiley IntesciencePulications, 2001.								
Reference Books:								
1. Mechanics & Thermodynamics of Propulsion 2. M. Barrere, A. Jaumotte, B.J. Veubeke and J. Vanderkerckhove, Rocket Propulsion, Elsevier Publishing Company, Amsterdam, 1960 (remarks: no reference books available)								

AEROSPACE STRUCTURES

IV Semester:								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AE10	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
UNIT-I	THIN PLATE THEORY, STRUCTURAL INSTABILITY							
Analysis of thin rectangular plates subject to bending, twisting, distributed transverse load, combined bending and in-plane loading-thin plates having small initial curvature,energy methods of analysis.								
UNIT-II	BUCKLING OF THIN PLATES AND BENDING OF THIN-WALLED BEAMS							
Buckling of thin plates-elastic, inelastic, experimental determination of critical load for a flat plate, local instability, Tension field beams- complete diagonal tension, incomplete diagonal tension. Unsymmetrical bending- resolution of bending moments, direct stress distribution, position of neutral axis. Deflections due to bending-approximations for thin walled sections, temperature effects.								
UNIT-III	SHEAR AND TORSION OF THIN-WALLED BEAMS							
Shear loaded thin Walled beams-general stress, strain and displacement relationships, direct stress, shear centre, twist and warping. Bending, shear, torsion of combined open and closed section beams.								
UNIT-IV	STRUCTURAL IDEALIZATION							
Structural idealization-principal assumptions, idealization of panel, effect on the analysis of thin Walled beams Under bending, shear, and torsion loading-application to determining deflection.								
UNIT-V	STRESS ANALYSIS OF AIRCRAFT COMPONENTS- WING and FUSELAGE							
Wing spars and box beams-tapered wing spar, beams having variable Stringer areas. Wings-Three-boom shell in bending, torsion, shear, tapered wings, deflections, cut-outs in wings. Bending, shear, torsion, cut-outs in fuselages, fuselage frames and wing ribs-principles of stiffener/ web construction, wing ribs.								
Text Books:								
1. Megson T. H. G (2012), Aircraft Structures for Engineering Students, 5 th edition, Elsevier, New York. 2. 3E F Bruhn (1973), Analysis and Design of Flight Vehicle Structures, Tri-state Offset Company, USA								
Reference Books:								
1. B. C. Punmia (2011), Theory of Structures, 13 th edition, Laxmi Publication, Hyderabad. 2. Timoshenko, Mechanics of Materials, CBS Publication								

PROBABILITY, STATISTICS AND COMPLEX

III Semester: MECH , AERO

Course Code:		Category		Hours / Week		Credits	Maximum Marks			
A6BS05		BSC		L	T	P	C	CIA	SEE	Total
				3	1	-	4	40	60	100
UNIT-I	RANDOM VARIABLES AND PROBABILITY DISTRIBUTIONS									
Random Variables – Discrete and Continuous. Probability distributions, mass function/ density function of a probability distribution, mathematical expectation,. Binomial, Poisson, Normal distributions, and their Properties and hence find mean and variance										
UNIT-II	TESTING OF HYPOTHESIS - I									
Testing of hypothesis: Null hypothesis, Alternate hypothesis, Type I& Type II errors – critical region, confidence interval, Level of significance. One sided test, Two sided test. Large sample tests: (i) Test of Equality of means of two samples, equality of sample mean and population mean (cases of known variance & unknown variance, equal and unequal variances) (ii) Tests of significance of difference between sample S.D and population S.D. (iii) Tests of significance difference between sample proportion and population proportion, difference between two sample proportions.										
UNIT-III	TESTING OF HYPOTHESIS-II									
Student t-distribution, its properties; Test of significance sample mean and population mean, difference between means of two small samples. Snedecor's F- distribution and its properties. Test of equality of two population variances. Chi-square distribution, its properties, Chi-square test of goodness of fit.										
UNIT-IV	COMPLEX DIFFERENTIATION									
Limit, Continuity and Differentiation of Complex functions, Analyticity, Cauchy-Riemann equations (without proof), finding harmonic conjugate.										
UNIT-V	COMPLEX INTEGRATION									
Line integral, Cauchy's theorem, Cauchy's Integral formula, Zeros of analytic functions, Singularities, Taylor's series, Laurent's series, Residues, Cauchy Residue theorem (All theorems without Proof).										
Text Books:										
1. B.S.Grewal, Higher Engineering Mathematics, Khanna publishers, 36th Edition, 2010. 2. Probability and Statistics for Engineers by Richard Arnold Johnson, Irwin Miller and John E. Freund, New Delhi, Prentice Hall. 3.Probability and Statistics for Engineers and Sciences by Jay L. Devore, Cengage Learning										
Reference Books:										
1. Ervin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006. 2. Fundamentals of Mathematical Statistics by S.C. Guptha&V.K. Kapoor, S. Chand 3. Introduction to Probability and Statistics for Engineers and Scientists by Sheldon M. Ross, Academic Press										

FLIGHT SIMULATION AND PROPULSION LAB

IV Semester								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AE11	PCC	L	T	P	C	CIA	SEE	Total
		-	-	3	1.5	40	60	100
LIST OF EXPERIMENTS								
PROPULSION LAB 1. Stripping of a piston engine, visual inspection and reasoning for common troubles and trouble shooting 2. Study of piston engine (Port Timing Diagram) 3. Study of piston engine (Valve Timing) 4. Performance of piston engine 2-stroke 5. Performance of piston engine 4-stroke 6. Engine Balancing.								
FLIGHT SIMULATION LAB 1. Demonstration of cockpit display systems and simulator commands 2. Airplane glide performance 3. Calculation of CL and CD using the Speed-Power Method 4. Effect of Flaps on Take-off and landing Performance 5. Effect of Weight on Take-off and landing Performance 6. Estimate the Distance travelled by the aircraft using GPS								
Reference Books: 1. H. Cohen, G. F. C. Rogers and H. I. H. Saravanamuttoo, "Gas Turbine Theory", Longman, 2006. 2. M. L. Mathur and R. P. Sharma, "Gas Turbine, Jet and Rocket Propulsion", Standard Publishers & Distributors, Delhi, 2007.								

IOT LAB**IV Semester**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AE61	ESC	L	T	P	C	CIA	SEE	Total
		-	-	2	1	40	60	100

LIST OF EXPERIMENTS**Introduction to IOT**

1. Introduction to basic electronic components and digital electronic
2. Introduction to sensors and Actuators
3. Introduction to microcontroller
4. Introduction to Arduino IDE

EXPERIMENT PROGRAMS:

1. Blinking of LED with different delays
2. Digital I/O Interface [IR Sensor, PIR Sensor]
3. Analog Interface [ADC, Temperature Sensor]
4. Motor speed And Direction control
5. Serial Communication
6. Wireless Interface –Bluetooth & Wi-Fi Technologies
7. Wireless Control of wheeled robot
8. Smart Home Android App Development

Reference Books:

1. Sylvia Libow Martinez, Gary S Stager, Invent To Learn: Making, Tinkering, and Engineering in the Classroom, Constructing Modern Knowledge Press, 2016
2. Michael Margolis, Arduino Cookbook, Oreilly, 2011

AIRCRAFT INTERIOR DESIGN LAB**IV Semester**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AE12	PCC	L	T	P	C	CIA	SEE	Total
		-	-	3	1.5	30	70	100

COURSE OBJECTIVES:

1. To design various Aircraft interior components in lab.
2. To design various individual components, sub-assemblies and main assemblies in lab.
3. Discuss the importance of design process and studying the different phases of designing process involved in the design.
4. Understand the design of aircraft concepts.
5. To design various aircraft components by using Catia software

LIST OF EXPERIMENTS

- 1 Aircraft Seating Design Arrangements (General Class & Business Class) (2D)
- 2 Aircraft Single Aisle Design(3D & 2D)
- 3 Aircraft Twin Aisle Design(3D & 2D)
- 4 Aircraft Seat Design (3D- Business Class)
- 5 Aircraft Seat Design (3D- First Class)
- 6 Design of Laptop Tray Sizing for Seating in Civil Aircrafts.
- 7 Design of Window for Civil Aircraft
- 8 Design of Wind Shield for Fighter Aircraft(Select any one airplane)
- 9 Design of Wind shield for Passenger Aircraft(Select any one Airplane)
- 10 Design of Arm rest cap height from surface in Civil Aircraft
- 11 Design of Joystick(Spacing) in Fighter Aircraft

Note: Ten experiments should be performed.

Reference Books:

Basics in Catia V5 with Simulation by Ranjan Chikesh

COURSE OUTCOMES:

1. Understand the importance of drawing and design process and phases involved in the design process
2. Design various individual components, sub-assemblies and main assemblies in design lab
3. Design various orthographic and isometric projections in drawing sheets
4. Develop the basic concepts of aircraft interiors
5. Design and develop aircraft interior components

PYTHON COMPUTING LAB

IV Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AE13	ESC	L	T	P	C	CIA	SEE	Total
		-	--	2	1	40	60	100

LIST OF PYTHON PROGRAMS

1. Write a program to demonstrate different number data types in Python.
 2. Write a program to perform different Arithmetic Operations on numbers in Python.
 3. Write a program to create, concatenate and print a string and accessing sub-string from a given string.
 4. Write a python script to print the current date in the following format "Sun Oct 30 02:26:23 IST 2022"
 5. Write a program to create, append, and remove lists in python.
 6. Write a program to demonstrate working with tuples in python.
 7. Write a program to demonstrate working with dictionaries in python.
 8. Write a python program to find largest of three numbers.
 8. Write a program to perform equations of uniform motion of kinematics :
 - i. $v = u + at$
 - ii. $s = ut + \frac{1}{2}(at^2)$
 - iii. $v^2 = u^2 - 2as$
 9. Write the menu program to find the relationship between stress and strain curve as given below:
 - i. Young's Modulus
 - ii. Shear Modulus
 - iii. Poisson Ratio
 10. Write the program to determine the shear force and bending moment in beams.
 11. Write a program to calculate total pressure exerted in ideal fluid as equation is given below:
 $P + \frac{1}{2}(\rho v^2) + \rho gh = \text{constant}$
 Where P is Pressure, V is Velocity of fluid, ρ is density and h is the height of the container.
 12. Write a program to compute eigen value and vector of a given 3*3 matrix using NumPy.
- Note:** Minimum 12 experiments are to be conducted.

Reference Books:

1. Reema Thareja, Python Programming: Using Problem Solving Approach, Oxford University Press.
2. Lutz M, Learning Python, Shroff

GENDER SENSITIZATION

IV Semester								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6HS05	HSME	L	T	P	C	CIE	SEE	Total
		2	--	--	Nil	40	60	100
UNIT-I								
UNDERSTANDING GENDER: Gender: Why Should We Study It? (<i>Towards a World of Equals</i> : Unit -1) Socialization: Making Women, Making Men (<i>Towards a World of Equals</i> : Unit -2) Introduction. Preparing for Womanhood. Growing up Male. First lessons in Caste. Different Masculinities.								
UNIT-II								
GENDER AND BIOLOGY: Missing Women: Sex Selection and Its Consequences (<i>Towards a World of Equals</i> : Unit -4) Declining Sex Ratio. Demographic Consequences. Gender Spectrum: Beyond the Binary (<i>Towards a World of Equals</i> : Unit -10) Two or Many? Struggles with Discrimination								
UNIT-III								
GENDER AND LABOUR: Housework: the Invisible Labour (<i>Towards a World of Equals</i> : Unit -3) "My Mother doesn't Work." "Share the Load." Women's Work: Its Politics and Economics (<i>Towards a World of Equals</i> : Unit -7) Fact and Fiction. Unrecognized and Unaccounted work. Additional Reading: Wages and Conditions of Work.								
UNIT-IV								
ISSUES OF VIOLENCE: Sexual Harassment: Say No! (<i>Towards a World of Equals</i> : Unit -6) Sexual Harassment, not Eve-teasing- Coping with Everyday Harassment- Further Reading: "Chupulu". Domestic Violence: Speaking Out (<i>Towards a World of Equals</i> : Unit -8) Is Home a Safe Place? -When Women Unite [Film]. Rebuilding Lives. Additional Reading: New Forums for Justice. Thinking about Sexual Violence (<i>Towards a World of Equals</i> : Unit -11) Blaming the Victim-"I Fought for my Life...." - Additional Reading: The Caste Violence.								
UNIT-V								
GENDER: CO – EXISTENCE: Just Relationships: Being Together as Equals (<i>Towards a World of Equals</i> : Unit -12) Mary Kom and Onler. Love and Acid just do not Mix. Love Letters. Mothers and Fathers Additional Reading: Rosa Parks-The Brave Heart								
TEXT BOOKS:								
1. All the five Units in the Textbook, " <i>Towards a World of Equals: A Bilingual Textbook on Gender</i> " written by A. Suneetha, Uma Bhugubanda, Duggirala Vasanta, Rama Melkote, Vasudha Nagaraj, Asma Rasheed, Gogu Shyamala, Deepa Sreenivas and Susie Tharu and published by Telugu Akademi, Hyderabad, Telangana State in the year 2015 .								
REFERENCE BOOKS:								
1. Menon, Nivedita. Seeing like a Feminist. New Delhi: Zubaan-Penguin Books, 2012 2. Abdulali Sohaila. " <i>I Fought For My Life...and Won</i> ." Available online at: http://www.thealternative.in/lifestyle/i-fought-for-my-lifeand-won-sohaila-abdulali/								